

Stochastic Processes Study Guide

Course: Math 311-3

Name: Moinak Nath

Date: June 11, 2025

Chapter 0: Background Knowledge and Review

We begin with some basic definitions from measure theory. Although this class is not technically measure theory based, it is sufficiently advanced as to flirt with it, and the professor keeps constantly mentioning it so...

1) Measure Theoretic Probability Framework

Definition (Sigma-algebra and measurable space)

Let Ω be a set. This set will be our "universe." A collection $\mathcal{F}$ of subsets of Ω is called a σ -algebra (or σ -field) if the following three properties hold:

1. $\Omega \in \mathcal{F}$
2. If $E \in \mathcal{F}$, then $E^c \in \mathcal{F}$
3. $\mathcal{F}$ is closed under countable union. That is, if $E_k \in \mathcal{F}, k \geq 1$, then

$$\bigcup_{k=1}^{\infty} E_k \in \mathcal{F}$$

If $\mathcal{F}$ is a σ -algebra on Ω , then we call the pair $(\Omega, \mathcal{F})$ a *measurable space* and any element $E \in \mathcal{F}$ a *measurable set*.

Note. (b) and (c), together with de Morgan's laws, imply that a σ -algebra $\mathcal{F}$ is also closed under countable intersections: If $E_k \in \mathcal{F}$ for $k \geq 1$, then

$$\bigcap_{k \geq 1} E_k \in \mathcal{F}$$

Definition (Sigma algebra generated by a collection of sets)

Let Ω be a set and $\mathcal{A}$ a collection of subsets of Ω . The σ -algebra generated by $\mathcal{A}$ is:

$$\sigma(\mathcal{A}) := \bigcap \{ \mathcal{F} : \mathcal{F} \text{ is a } \sigma\text{-algebra on } \Omega \text{ and } \mathcal{A} \subseteq \mathcal{F} \}$$

Definition (Measure)

A **measure** is a nonnegative countably additive set function. In other words, a function $\mu : \mathcal{F} \rightarrow \mathbb{R}$ with:

1. $\mu(A) \geq \mu(\emptyset) = 0$
2. If $A_i \in \mathcal{F}$ is a countable sequence of disjoint sets, then $\mu(\bigcup A_i) = \sum_{i=0}^{\infty} \mu(A_i)$

Note. An obtuse mathematical definition which means, in words, that "the sigma algebra generated by $\mathcal{A}$ is the smallest (by cardinality) sigma algebra on Ω containing $\mathcal{A}$."

Definition (Probability measure and probability space)

Let $(\Omega, \mathcal{F})$ be a measurable space and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ a function for which:

- $\mathbb{P}(\Omega) = 1$
- $\mathbb{P}$ has the σ -additivity property. That is, if $E_k, k \in \mathbb{N}$, are pairwise disjoint sets of $\mathcal{F}$, then:

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(E_k)$$

then $\mathbb{P}$ is called a *probability measure* on the measurable space $(\Omega, \mathcal{F})$. The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a *probability space*. The set Ω is the **sample space** (set of all possible outcomes) and $\mathcal{F}$ is **underlying sigma algebra**

Definition (Event)

For a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, the elements of $\mathcal{F}$ are called **events**

Note. Events are exactly the sets of outcomes (subsets of Ω) for which a probability (measure of likelihood) is defined. This formalizes the notion of what it means to write "probability of this happening is x". Also, notice that we need the sigma algebra and the

probability measure in order for the probability space to behave in a nice way (i.e. be useful for practical purposes)

Definition (Sample Points ω)

Elements $\omega \in \Omega$ of the sample space Ω in a probability space are called **sample points**

Definition (Almost Sure/With Probability 1)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We say an event $E \in \mathcal{F}$ happens almost surely if $\mathbb{P}(E) = 1$

Definition (Continuity of Probability)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $E_k, k \geq 1$ a sequence of events:

1. If $E_1 \subseteq E_2 \subseteq \dots$, then:

$$\lim_{k \rightarrow \infty} \uparrow \mathbb{P}(E_k) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} E_k\right)$$

2. If $E_1 \supseteq E_2 \supseteq \dots$, then:

$$\lim_{k \rightarrow \infty} \downarrow \mathbb{P}(E_k) = \mathbb{P}\left(\bigcap_{k=1}^{\infty} E_k\right)$$

Remark

In the previous definition, 1 is known as **continuity from below**, while 2 is **continuity from above**

Definition (Random Variable)

A function $X : \Omega \rightarrow \mathbb{R}$ is a random variable if for every Borel set $B \subset \mathbb{R}$ the preimage of B , $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$.

Definition ($\mathcal{F}$ -measurable)

A random variable X is $\mathcal{F}$ -measurable (we may write $X \in \mathcal{F}$)

$$\mathbb{1}_{\omega \in A}$$

Definition (Indicator Function)

Let $A \in \mathcal{F}$ be an event in a probability space. The *indicator function* of A , denoted by $\mathbb{1}_A$, is defined as:

$$\mathbb{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Definition (σ -algebra generated by a random variable)

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The *σ -algebra generated by X* , denoted $\sigma(X)$, is the collection of all subsets of Ω of the form

$$\sigma(X) = \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\},$$

where $\mathcal{B}(\mathbb{R})$ denotes the Borel σ -algebra on $\mathbb{R}$.

In other words, $\sigma(X)$ is the smallest σ -algebra with respect to which X is measurable.

Note. I will not bother to define Borel sets or Borel Sigma Algebra, but I think of them as the nice sets on $\mathbb{R}$ and the Borel σ -algebra is the collection of all Borel sets on $\mathbb{R}$. The key here is that for a random variable $\sigma(X)$ means the σ -algebra generated by the preimages of the Borel sets, under X .

Interpretation: Continuous time Markov chains theory requires knowledge of how to solve systems of ODEs.

2) Analysis Preliminaries

a) Supremum and Infimum

Definition (Supremum and Infimum)

Let $S \subseteq \mathbb{R}$ be a nonempty set.

Supremum (least upper bound): $\sup S$ is the smallest number M such that:

- (a) $x \leq M$ for all $x \in S$ (upper bound)
- (b) If M' is any upper bound of S , then $M \leq M'$ (least)

Infimum (greatest lower bound): $\inf S$ is the largest number m such that:

- (a) $x \geq m$ for all $x \in S$ (lower bound)
- (b) If m' is any lower bound of S , then $m \geq m'$ (greatest)

Conventions: $\sup \emptyset = -\infty$, $\inf \emptyset = +\infty$, $\sup S = +\infty$ if S is unbounded above, $\inf S = -\infty$ if S is unbounded below.

Remark (Supremum vs Maximum)

- $\sup S$ always exists in $\mathbb{R} \cup \{-\infty, +\infty\}$
- $\max S$ exists only if $\sup S \in S$ and S achieves its supremum
- Similarly for $\inf S$ vs $\min S$

b) Limit Superior and Limit Inferior

Definition (Limit Superior and Limit Inferior)

For a sequence $(a_n)_{n \geq 1}$ of real numbers:

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k = \inf_{n \geq 1} \sup_{k \geq n} a_k$$

$$\liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k = \sup_{n \geq 1} \inf_{k \geq n} a_k$$

Proposition (Properties of Limsup and Liminf)

- (a) $\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n$
- (b) $\lim_{n \rightarrow \infty} a_n$ exists if and only if $\liminf_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n$
- (c) If the limit exists: $\lim_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n$
- (d) $\limsup_{n \rightarrow \infty} (-a_n) = -\liminf_{n \rightarrow \infty} a_n$

Definition (Limsup and Liminf for Sets)

For a sequence of sets $(A_n)_{n \geq 1}$:

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \{\omega : \omega \in A_n \text{ for infinitely many } n\}$$

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k = \{\omega : \omega \in A_n \text{ for all but finitely many } n\}$$

3) Borel-Cantelli Lemma

Lemma (Borel-Cantelli Lemma)

Let $(A_n)_{n \geq 1}$ be a sequence of events in a probability space.

First Borel-Cantelli Lemma: If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then:

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) = \mathbb{P}(A_n \text{ occurs infinitely often}) = 0$$

Second Borel-Cantelli Lemma: If the events (A_n) are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then:

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) = \mathbb{P}(A_n \text{ occurs infinitely often}) = 1$$

Remark (Borel-Cantelli Interpretation)

The Borel-Cantelli lemma provides a criterion for almost sure convergence:

- If the sum of probabilities converges, then only finitely many events occur (almost surely)
- If the events are independent and the sum diverges, then infinitely many events occur (almost surely)

This is fundamental for proving strong laws of large numbers and almost sure convergence results.

4) Taylor Series and Geometric Series

Proposition (Common Taylor Series)

- (a) $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$
- (b) $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ for $|x| < 1$
- (c) $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$ for $|x| < 1$
- (d) $(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$ for $|x| < 1$

Proposition (Geometric Series)

- (a) $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ for $|r| < 1$
- (b) $\sum_{n=0}^N r^n = \frac{1-r^{N+1}}{1-r}$ for $r \neq 1$
- (c) $\sum_{n=1}^{\infty} n r^{n-1} = \frac{1}{(1-r)^2}$ for $|r| < 1$
- (d) $\sum_{n=0}^{\infty} (n+1) r^n = \frac{1}{(1-r)^2}$ for $|r| < 1$

Convergence Tests

Theorem (p-Series Test)

For Series: The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

For Integrals: The improper integral $\int_1^{\infty} \frac{1}{x^p} dx$ converges if and only if $p > 1$.

General Form:

- $\sum_{n=a}^{\infty} \frac{1}{n^p}$ converges $\iff p > 1$ (for any $a \geq 1$)
- $\int_a^{\infty} \frac{1}{x^p} dx$ converges $\iff p > 1$ (for any $a > 0$)

Theorem (Ratio Test)

Let $\sum_{n=1}^{\infty} a_n$ be a series with $a_n > 0$. Define:

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$$

Then:

- (a) If $L < 1$, the series converges
- (b) If $L > 1$, the series diverges
- (c) If $L = 1$, the test is inconclusive

Root Test Version: If $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$, the same conclusions hold.

Remark (Applications in Probability)

These tests are frequently used for:

- **MGF convergence:** Checking when $\mathbb{E}(e^{tX}) = \sum_{n=0}^{\infty} \frac{t^n \mathbb{E}(X^n)}{n!}$ converges
- **Power series:** Finding radius of convergence for generating functions
- **Tail bounds:** Determining when $\sum_{n=1}^{\infty} \mathbb{P}(X \geq n)$ converges (related to finiteness of moments)
- **Renewal theory:** Convergence of renewal equations and Laplace transforms

Remark (Quick Reference)

Common convergent series:

- $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$ (converges)
- $\sum \frac{1}{n \ln n}$ (diverges)
- $\sum \frac{1}{n(\ln n)^2}$ (converges)

Ratio test examples:

- $\sum \frac{x^n}{n!}$: $L = 0 < 1$ for any x (converges for all x)
- $\sum n!x^n$: $L = \infty > 1$ for $x \neq 0$ (diverges for $x \neq 0$)

5) Matrix Operations (2×2 Case)

Proposition (2×2 Matrix Formulas)

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

Determinant: $\det(A) = ad - bc$

Inverse: If $\det(A) \neq 0$:

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Eigenvalues: Solutions to $\det(A - \lambda I) = 0$:

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

$$\lambda = \frac{(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}}{2}$$

Trace: $\text{tr}(A) = a + d$ (sum of eigenvalues)

6) First-Order Ordinary Differential Equations

First order ODEs are trivial and if you don't know them you are stupid, but:

Remark

A first order linear ODE of the form $a_0(x) + y'(x) + a_1(x)y(x) = f(x)$ may be easily solved using the integrating factor:

$$p(x) = e^{\int \frac{a_1}{a_0} dx}$$

by multiplying the equation by $p(x)$ and noticing the product rule on the LHS

Proof. A trivial exercise in basic calculus. □

a) Eigenvalue Methods

Remark (Applications to Differential Equations)

For the system $\frac{d\vec{x}}{dt} = A\vec{x}$, the solution is $\vec{x}(t) = e^{At}\vec{x}(0)$ where:

- If A has distinct eigenvalues λ_1, λ_2 with eigenvectors $\vec{v}_1, \vec{v}_2$
- Then $e^{At} = c_1 e^{\lambda_1 t} \vec{v}_1 \vec{v}_1^T + c_2 e^{\lambda_2 t} \vec{v}_2 \vec{v}_2^T$ (in appropriate coordinates)
- This is crucial for solving birth-death chain differential equations in CTMCs

Definition (Little-oh Notation)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. The function f is said to be $o(h)$ if

$$\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0$$

Intuitively, this means that $f(h)$ is asymptotically strictly smaller than $h, h \rightarrow 0$

Chapter 1: Probability

This chapter will provide a rapid run through of everything that would be covered in a first course in probability.

Proposition (Probability Properties)

Let $E, E_1, E_2 \in \mathcal{F}$ be events. Then the definition of probability forces the following properties hold:

1. $\mathbb{P}(\emptyset) = 0$
2. $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$
3. $\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2) - \mathbb{P}(E_1 \cap E_2)$
4. If $E_1 \subseteq E_2 \implies \mathbb{P}(E_1) \leq \mathbb{P}(E_2)$

Theorem (Union Bound and Inclusion-Exclusion Principle)

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mathbb{P}(E_i)$$

In particular, for finite collections:

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^n E_i\right) &= \sum_{i=1}^n \mathbb{P}(E_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(E_i \cap E_j) + \sum_{1 \leq i < j < k \leq n} \mathbb{P}(E_i \cap E_j \cap E_k) \\ &\quad - \dots + (-1)^{n+1} \mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_n) \end{aligned} \quad (1)$$

Special cases:

- For two events: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- For three events:

$$\begin{aligned} \mathbb{P}(A \cup B \cup C) &= \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) \\ &\quad - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C) \end{aligned} \quad (2)$$

- If events are disjoint: $\mathbb{P}(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n \mathbb{P}(E_i)$ (equality holds)

Definition (Binomial and Multinomial Coefficients)

The *binomial coefficient* is defined as

$$\binom{n}{r} = \frac{n!}{r!(n-r)!},$$

and it gives the number of ways to choose r elements from a set of n elements.

The *multinomial coefficient* generalizes this to multiple groups. If $n = k_1 + k_2 + \dots + k_m$, then the multinomial coefficient is

$$\binom{n}{k_1, k_2, \dots, k_m} = \frac{n!}{k_1! k_2! \dots k_m!},$$

and it represents the number of ways to partition n items into m groups of sizes $k_1, k_2, \dots, k_m$.

Theorem (Binomial and Multinomial Theorem)

(Binomial Theorem) For any integer $n \geq 0$ and any real (or complex) numbers x and y , we have

$$(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}.$$

(Multinomial Theorem) More generally, for any integer $n \geq 0$ and any real (or complex) numbers $x_1, x_2, \dots, x_m$, we have

$$(x_1 + x_2 + \dots + x_m)^n = \sum_{k_1 + k_2 + \dots + k_m = n} \binom{n}{k_1, k_2, \dots, k_m} x_1^{k_1} x_2^{k_2} \dots x_m^{k_m},$$

where the sum is over all nonnegative integers $k_1, k_2, \dots, k_m$ such that $k_1 + k_2 + \dots + k_m = n$.

Definition (Conditional Probability)

Given events $A, B \in \mathcal{F}$ we have the *conditional probability* of A given B , denoted $\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$

Remark (Conditional Probability is a Probability)

Conditional probability is a probability measure and the same rules apply as regular probability as long as you keep writing $| B$!

Note.

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A) \mathbb{P}(B | A) \mathbb{P}(C | A \cap B)$$

This is a direct application of the **chain rule of probability**, which generalizes to any finite collection of events $A_1, A_2, \dots, A_n$ as:

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \mathbb{P}(A_1) \prod_{i=2}^n \mathbb{P}\left(A_i \mid \bigcap_{j=1}^{i-1} A_j\right).$$

Proposition (Law of Total Probability (Measure-Theoretic Form))

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. In particular, let $A_i \subset \mathcal{F}, i \geq 1$ be a finite or countable partition of Ω such that $\mathbb{P}(A_i) > 0$ for all i . Then for any event $B \in \mathcal{F}$,

$$\mathbb{P}(B) = \sum_{i=1}^{\infty} \mathbb{P}(B | A_i) \mathbb{P}(A_i).$$

Definition (Independence of Events)

Let $A, B \in \mathcal{F}$. We say that A and B are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B).$$

Proposition

Let $A, B \in \mathcal{F}$ be independent events. Then:

1. A is independent of B^c : $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) \mathbb{P}(B^c)$,
2. A^c is independent of B ,
3. A^c is independent of B^c .

Definition (Mutual Independence)

Let $A_1, A_2, \dots, A_n \in \mathcal{F}$. These events are said to be *mutually independent* if for every subset $\{i_1, i_2, \dots, i_k\} \subseteq \{1, 2, \dots, n\}$, we have

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_k}),$$

for all $1 \leq k \leq n$.

Meaning: Pick any of them, must be independent

Theorem (Bayes' Theorem)

Let $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. Then

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A) \mathbb{P}(A)}{\mathbb{P}(B)}.$$

Probability Functions and Properties

A random variable taking values in a set $\{X \in S\}$ is an event.

Definition (Probability Mass Function)

For a discrete random variable X , the Probability Mass Function (PMF) is:

$$p_X(x) = \mathbb{P}(X = x)$$

where:

$$\sum_x p_X(x) = 1$$

In parallel, for a continuous random variable, we have the probability density function.

Definition (Probability Density Function)

For a continuous random variable X , the Probability Density Function (PDF) $f_X(x)$ is defined by:

$$f_X(x) = \lim_{h \rightarrow 0^+} \frac{\mathbb{P}(x < X \leq x + h)}{h}$$

provided this limit exists (in this class, it always will!)

Proposition (PDF Properties)

- (a) For $B \subseteq \text{range}(X)$, the probability $\mathbb{P}(X \in B) = \int_B f_X(x) dx$
- (b) $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- (c) $\mathbb{P}(X = a) = 0$ for any specific value a

Equivalently: $f_X(x) = \frac{d}{dx} F_X(x)$ where F_X is differentiable.

Properties:

- (a) $f_X(x) \geq 0$ for all x
- (b) $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- (c) $F_X(x) = \int_{-\infty}^x f_X(t) dt$
- (d) $\mathbb{P}(X = a) = 0$ for any specific value a

Remark (PDF Interpretation)

The PDF is *not* a probability. Rather, integrating mass density $\rho(\vec{x})$ over a region gives you a probability, just as mass $m = \int_R \rho(\vec{x}) dV$

Definition (Cumulative Distribution Function (CDF))

For any random variable X :

$$F_X(x) = \mathbb{P}(X \leq x)$$

Discrete Case:

$$F_X(a) = \sum_{x \leq a} p_X(x)$$

Continuous Case: If X has PDF $f_X(t)$, then:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Proposition (CDF Properties)

Let F_X be the CDF of random variable X . Then:

Universal Properties:

- (a) F_X is non-decreasing: if $x_1 < x_2$, then $F_X(x_1) \leq F_X(x_2)$
- (b) $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$
- (c) F_X is right-continuous: $\lim_{h \rightarrow 0^+} F_X(x+h) = F_X(x)$ for all x

Continuous Case Properties:

- (d) F_X is continuous everywhere
- (e) $f_X(x) = \frac{d}{dx}[F_X(x)]$
- (f) $\mathbb{P}(a < X < b) = F_X(b) - F_X(a)$ (could be open or closed on right or left)

Discrete Case Properties:

- (g) F_X has jump discontinuities at points where $\mathbb{P}(X = x_i) > 0$
- (h) F_X is piecewise constant between jump points (closed on left open at right)
- (i) Jump size at x_i : $\mathbb{P}(X = x_i) = F_X(x_i) - F_X(x_i^-)$

Continuity Characterization:

- (j) X is continuous $\iff F_X$ is continuous everywhere
- (k) X is discrete $\iff F_X$ is a step function with countably many jumps
- (l) F_X is continuous at a $\iff \mathbb{P}(X = a) = 0$

Proposition (Discrete CDF Probability Calculations)

- (a) $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$
- (b) $\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a^-)$
- (c) $\mathbb{P}(a < X < b) = F_X(b^-) - F_X(a)$
- (d) $\mathbb{P}(a \leq X < b) = F_X(b^-) - F_X(a^-)$

Remark (Survival Function)

The survival function (or reliability function) is: $S_X(x) = \mathbb{P}(X > x) = 1 - F_X(x)$

Properties:

- (a) S_X is non-increasing
- (b) $\lim_{x \rightarrow -\infty} S_X(x) = 1$ and $\lim_{x \rightarrow \infty} S_X(x) = 0$
- (c) For continuous X : $f_X(x) = -\frac{d}{dx} S_X(x)$

Theorem (Functions of Random Variables - CDF Method)

Let X be a continuous random variable with PDF $f_X(x)$ and CDF $F_X(x)$. Let $Y = g(X)$ where g is a function. Then the CDF of Y is given by:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y)$$

If g is monotonic and differentiable, then:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Remark (Non-monotonic Example)

For $Y = X^2$ where X has PDF $f_X(x)$:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^2 \leq y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

Therefore:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}} = \frac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}}$$

for $y > 0$, and $f_Y(y) = 0$ for $y \leq 0$.

The takeaway: If $Y = g(X)$, we can often find an expression by solving for the region in terms of X . Like $\mathbb{P}(Y \in A) = \mathbb{P}(g(X) \in A) = \mathbb{P}(X \in g^{-1}(A)) = \mathbb{P}(X \in B)$

Multivariate Distributions

Definition (Joint CDF)

For random variables X and Y , the joint cumulative distribution function is:

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y)$$

Discrete Case

Definition (Joint PMF)

For discrete random variables X and Y , the joint probability mass function is:

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$$

Proposition (Discrete Joint PMF Properties)

Properties:

- (a) $\sum_x \sum_y p_{X,Y}(x, y) = 1$
- (b) $F_{X,Y}(x, y) = \sum_{s \leq x} \sum_{t \leq y} p_{X,Y}(s, t)$

Continuous Case

Definition (Joint PDF - Limit Definition)

For continuous random variables X and Y , the joint probability density function $f_{X,Y}(x, y)$ is defined by:

$$f_{X,Y}(x, y) = \lim_{h_1, h_2 \rightarrow 0^+} \frac{\mathbb{P}(x < X \leq x + h_1, y < Y \leq y + h_2)}{h_1 h_2}$$

provided this limit exists.

Proposition (Jointly Continuous PDF Properties)

For the joint PDF $f_{X,Y}(x, y)$:

- (a) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$
- (b) $F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(s, t) dt ds$
- (c) $f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y)$
- (d) $\mathbb{P}((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$ for any region A

Theorem (Change of Variables Formula)

Let (X, Y) have joint PDF $f_{X,Y}(x, y)$ and consider the transformation:

$$U = g_1(X, Y), \quad V = g_2(X, Y)$$

If the transformation is one-to-one with inverse:

$$X = h_1(U, V), \quad Y = h_2(U, V)$$

and the Jacobian $J = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \begin{vmatrix} \frac{\partial h_1}{\partial u} & \frac{\partial h_1}{\partial v} \\ \frac{\partial h_2}{\partial u} & \frac{\partial h_2}{\partial v} \end{vmatrix} \neq 0$

Then the joint PDF of (U, V) is:

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) \cdot |J|$$

Definition (Marginal Distributions)

From the joint distribution, we can obtain marginal distributions:

Discrete Case:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Continuous Case:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

Marginal CDFs:

$$F_X(x) = \lim_{y \rightarrow \infty} F_{X,Y}(x, y), \quad F_Y(y) = \lim_{x \rightarrow \infty} F_{X,Y}(x, y)$$

TLDR: Sum out the other variable

Definition (Independent Random Variables)

Random variables X and Y are independent $\iff$:

Using CDFs: $F_{X,Y}(x, y) = F_X(x)F_Y(y)$ for all (x, y)

Discrete Case: $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ for all (x, y)

Continuous Case: $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ for all (x, y)

Definition (Random Vector)

A random vector $\vec{X} = (X_1, X_2, \dots, X_n)$ is an n -dimensional vector whose components $X_1, X_2, \dots, X_n$ are random variables defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Equivalently, a random vector is a function $\vec{X} : \Omega \rightarrow \mathbb{R}^n$ such that each component $X_i : \Omega \rightarrow \mathbb{R}$ is a random variable.

The random vector $\vec{X}$ takes values in $\mathbb{R}^n$ and is characterized by its joint distribution.

Remark (Extension to Higher Dimensions)

All definitions extend naturally to n dimensions. For $(X_1, X_2, \dots, X_n)$:

Joint CDF: $F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n)$

Joint PMF (Discrete):

$$p_{X_1, \dots, X_n}(x_1, \dots, x_n) = \mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$$

Joint PDF (Continuous):

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \dots \partial x_n} [F_{X_1, \dots, X_n}(x_1, \dots, x_n)]$$

CDF from PMF/PDF:

- **Discrete:**

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \sum_{t_1 \leq x_1} \dots \sum_{t_n \leq x_n} p_{X_1, \dots, X_n}(t_1, \dots, t_n)$$

- **Continuous:**

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f_{X_1, \dots, X_n}(t_1, \dots, t_n) dt_n \dots dt_1$$

Change of Variables: For transformation $\vec{U} = \vec{g}(\vec{X})$, the joint PDF involves the $n \times n$ Jacobian determinant of the inverse transformation.

Independence: $X_1, \dots, X_n$ are independent $\iff$:

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = f_{X_1}(x_1) \dots f_{X_n}(x_n)$$

(or the analogous product form for PMFs and CDFs)

Convolution

Definition (Convolution)

Let X, Y be independent random variables with probability density functions f_X, f_Y respectively and let $Z = X + Y$. The probability density function of Z is the convolution of f_X and f_Y , denoted $f_X \star f_Y$:

$$f_Z(z) = (f \star g)(z) = \int_{-\infty}^{\infty} f_X(z - y)f_Y(y)dy = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)dx$$

Remark

Note that since $\frac{dF(x)}{dx} = f(x) \implies dF(x) = f(x)$, so we can write $dF(x)$ instead so convolution:

$$f_Z(z) = (f \star g)(z) = \int_{-\infty}^{\infty} f_X(z - y)dF(y) = \int_{-\infty}^{\infty} f_Y(z - x)dF(x)$$

An integral of the form

$$\int g(x)dF(x)$$

is known as Lebesgue–Stieltjes integral (but it's important that this is much more general than our focus, and in this case we don't assume the functions are nice: continuous differentiable, etc)

Expectation, Variance, and Covariance

Expectation (Expected Value)

Definition (Expectation)

For a random variable X , the expectation (or expected value) is:

$$\mathbb{E}(X) = \begin{cases} \sum_x x \cdot p_X(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x \cdot f_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Remark (Expectation's Existence)

$\mathbb{E}(X)$ exists $\iff \mathbb{E}(|X|) < \infty$.

Proposition (Fundamental Expectation Properties)

- (a) **Linearity:** $\mathbb{E}(a_1X_1 + \cdots + a_nX_n) = a_1\mathbb{E}(X_1) + \cdots + a_n\mathbb{E}(X_n)$ for constants a_i
- (b) **Constant:** $\mathbb{E}(c) = c$ for any constant $c \in \mathbb{R}$
- (c) **Monotonicity:** If $X \geq Y$ almost surely, then $\mathbb{E}(X) \geq \mathbb{E}(Y)$
- (d) **Independence:** If $X_1, \dots, X_n$ are independent, then $\mathbb{E}(X_1 \cdots X_n) = \mathbb{E}(X_1) \cdots \mathbb{E}(X_n)$

Theorem (Probability as Expectation)

For any event $A \in \mathcal{F}$:

$$\mathbb{P}(A) = \mathbb{E}(\mathbf{1}_A)$$

Theorem (Tail Sum Formula for Expectation)

For a non-negative random variable $X \geq 0$:

- (a) **Discrete:** $\mathbb{E}(X) = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k)$
- (b) **Continuous:** $\mathbb{E}(X) = \int_0^{\infty} \mathbb{P}(X > t) dt = \int_0^{\infty} S_X(t) dt$

Theorem (Law of the Unconscious Statistician (LOTUS))

For a function g and random variable X :

$$\mathbb{E}(g(X)) = \begin{cases} \sum_x g(x) \cdot p_X(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x) \cdot f_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Proposition (Multivariate Expectation)

For a random variable which is a function of two random variables $Z = g(X, Y)$, then:

$$\mathbb{E}(Z) = \int_{-\infty}^{\infty} z f_Z(z) dz = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} z f_{X,Y}(x, y) dx dy$$

Variance and Standard Deviation

Definition (Variance and Standard Deviation)

The variance of a random variable X is:

$$\text{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$$

The standard deviation is $\sigma_X = \sqrt{\text{Var}(X)}$.

Proposition (Variance Properties)

- (a) $\text{Var}(X) \geq 0$ with $\text{Var}(X) = 0 \iff X$ is constant
- (b) $\text{Var}(aX + b) = a^2 \text{Var}(X)$ for constants a, b
- (c) If X and Y are independent: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Covariance and Correlation

Definition (Covariance)

The covariance of random variables X and Y is:

$$\text{Cov}(X, Y) = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y))) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

General formula: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

Definition (Correlation)

The correlation coefficient of X and Y is:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

provided both variances are positive.

Proposition (Covariance Properties)

- (a) $\text{Cov}(X, X) = \text{Var}(X)$
- (b) $\text{Cov}(X, Y) = \text{Cov}(Y, X)$ (symmetry)
- (c) $\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z)$ (linear in both arguments)
- (d) If X and Y are independent, then $\text{Cov}(X, Y) = 0$ ($\Leftarrow$ DOES NOT HOLD)

Theorem (Variance of Sums)

For random variables $X_1, \dots, X_n$:

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

If $X_1, \dots, X_n$ are pairwise independent:

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i)$$

Advanced Properties and Formulas

Theorem (Law of Total Expectation (Tower Property))

For random variables X and Y :

$$\mathbb{E}(X) = \mathbb{E}(\mathbb{E}(X | Y))$$

This is the foundation of conditional expectation and states that we can compute expectations by conditioning on another variable.

Proposition (Sample Mean Properties)

Let $S_n = \sum_{i=1}^n X_i$ where $X_1, \dots, X_n$ are i.i.d. with $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Then:

- (a) $\mathbb{E}(S_n) = n\mu$ and $\text{Var}(S_n) = n\sigma^2$
- (b) For the sample mean $\bar{X}_n = \frac{S_n}{n}$: $\mathbb{E}(\bar{X}_n) = \mu$ and $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$

Moment Generating Functions

Definition (Moment Generating Function)

The moment generating function (MGF) of X is:

$$\phi_X(t) = \mathbb{E}(e^{tX})$$

provided this expectation exists for t in some neighborhood of 0.

Proposition (MGF Properties)

- (a) **Moments:** $\mathbb{E}(X^k) = \left. \frac{d^k}{dt^k} \phi_X(t) \right|_{t=0}$
- (b) **Uniqueness:** $\phi_X(t) = \phi_Y(t)$ for t near 0 $\iff X \stackrel{d}{=} Y$
- (c) **Independence:** If X and Y are independent, then $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$

Remark (Key Computational Insights)

- The tail sum formula is particularly useful in this class
- Probability is expectation: $\mathbb{P}(A) = \mathbb{E}(\mathbf{1}_A)$
- Covariance roughly speaking is a measure of correlation (i.e. dependence between X, Y), but $\text{Cov}(X, Y) = 0$ doesn't imply independence
- Independence is stronger than uncorrelatedness: Independent $\implies$ Uncorrelated, but not vice versa

Conditional Distributions

Conditional PMF and PDF

Definition (Conditional PMF)

For discrete random variables X and Y with $\mathbb{P}(Y = y) > 0$:

$$p_{X|Y}(x|y) = \mathbb{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}$$

Definition (Conditional PDF)

For continuous random variables X and Y with $f_Y(y) > 0$:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

Remark (Conditional Distributions are Distributions)

Once the conditional PMF or PDF is established, **all standard probability concepts apply** because conditional probability is still probability! Much like in events, same formulas as unconditional work as long as you state conditionality.

Conditional Expectation

Definition (Conditional Expectation)

Given the conditional distribution of X given $Y = y$:

Discrete case:

$$\mathbb{E}(X | Y = y) = \sum_x x \cdot p_{X|Y}(x|y)$$

Continuous case:

$$\mathbb{E}(X | Y = y) = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y) dx$$

The conditional expectation as a random variable is $\mathbb{E}(X | Y)$, which is a function of the random variable Y (i.e. $\mathbb{E}(X | Y) = g(Y)$)

Remark (Trick)

Rather than using the brute force formula in terms of the conditional distribution, it is often almost immediate what $\mathbb{E}(X | Y)$ is.

Tower Property

Theorem (Tower Property)

For random variables X and Y :

$$\mathbb{E}(X) = \mathbb{E}(\mathbb{E}(X | Y))$$

This fundamental property states that we can compute the expectation of X by:

- (a) First computing the conditional expectation $\mathbb{E}(X | Y = y)$ for each value of y
- (b) Then taking the expectation of this conditional expectation over Y

Explicit forms:

- **Discrete Y :**

$$\mathbb{E}(X) = \sum_y \mathbb{E}(X | Y = y) \mathbb{P}(Y = y)$$

- **Continuous Y :**

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} \mathbb{E}(X | Y = y) f_Y(y) dy$$

Theorem (Conditional Probability as Conditional Expectation)

For any event $A \in \mathcal{F}$ and random variable $Y \in \mathcal{F}$:

$$\mathbb{P}(A | Y) = \mathbb{E}(\mathbb{1}_A | Y)$$

Proposition (Properties of Conditional Expectation)

(a) **Taking out what is known:** If $g(Y)$ is a function of Y only:

$$\mathbb{E}(g(Y) \cdot X | Y) = g(Y) \cdot \mathbb{E}(X | Y)$$

(b) **Independence:** If X and Y are independent: $\mathbb{E}(X | Y) = \mathbb{E}(X)$

(c) **Iterated expectation:** For random variables X, Y, Z :

$$\mathbb{E}(X | Y) = \mathbb{E}(\mathbb{E}(X | Y, Z) | Y)$$

We can additionally condition on another RV Z , but must keep outer condition!

We will do an ever deeper dive into conditional expectation when we get to martingales. More than you ever wanted to know!

Probability Distributions Reference Table

Distribution	PMF/PDF	$\mathbb{E}[X]$	$\text{Var}(X)$	MGF
Discrete Distributions				
Bernoulli(p)	$p^k(1-p)^{1-k}$ $k \in \{0, 1\}$	p	$p(1-p)$	$1 - p + pe^t$
Binomial(n, p)	$\binom{n}{k}p^k(1-p)^{n-k}$ $k \in \{0, 1, \dots, n\}$	np	$np(1-p)$	$(1 - p + pe^t)^n$
Geometric(p)	$(1-p)^{k-1}p$ $k \in \{1, 2, 3, \dots\}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Geometric(p) (II)	$(1-p)^k p$ $k \in \{0, 1, 2, \dots\}$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$	$\frac{p}{1-(1-p)e^t}$
Negative Binomial(r, p)	$\binom{k-1}{r-1}p^r(1-p)^{k-r}$ $k \in \{r, r+1, \dots\}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$
Poisson(λ)	$\frac{e^{-\lambda}\lambda^k}{k!}$ $k \in \{0, 1, 2, \dots\}$	λ	λ	$e^{\lambda(e^t-1)}$
Continuous Distributions				

Distribution	PMF/PDF	$E[X]$	$\text{Var}(X)$	MGF
Uniform(a, b)	$\frac{1}{b-a}$ $x \in (a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
Exponential(λ)	$\lambda e^{-\lambda x}$ $x > 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ $x \in \mathbb{R}$	μ	σ^2	$e^{\mu t + \frac{\sigma^2 t^2}{2}}$
Gamma(α, β)	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ $x > 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\left(\frac{\beta}{\beta-t}\right)^\alpha$
Chi-Square(ν)	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}$ $x > 0$	ν	2ν	$(1-2t)^{-\nu/2}$
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$ $x \in (0, 1)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	No closed form

Remark (Distribution Significance)

- **Bernoulli:** Single trial success/failure
- **Binomial:** Sum of n i.i.d. Bernoulli trials
- **Geometric:** Number of trials until first success
- **Negative Binomial:** Number of trials until r -th success (sum of r i.i.d. Geometric)
- **Poisson:** Limit of Binomial as $n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda$; arrivals in Poisson process
- **Exponential:** Continuous memoryless distribution; interarrival times in Poisson process
- **Gamma:** Sum of α i.i.d. Exponential(β) random variables
- **Chi-Square:** Sum of squares of i.i.d. standard normals; special case of Gamma($\nu/2, 1/2$)
- **Beta:** Conjugate prior for Bernoulli/Binomial; ratio of Gamma random variables

Important Relationships

Remark (Key Distribution Relationships)

- Chi-Square(ν) is Gamma($\frac{\nu}{2}, \frac{1}{2}$)
- Exponential(λ) is Gamma($1, \lambda$)
- If $X \sim \text{Normal}(0, 1)$, then $X^2 \sim \chi^2(1)$
- If $X \sim \chi^2(m)$ and $Y \sim \chi^2(n)$ independently, then $\frac{X/m}{Y/n} \sim F(m, n)$
- If $X \sim \text{Normal}(0, 1)$ and $Y \sim \chi^2(\nu)$ independently, then $\frac{X}{\sqrt{Y/\nu}} \sim t(\nu)$
- Poisson(λ) approximates Binomial(n, p) when n large, p small, $np = \lambda$
- Normal(μ, σ^2) approximates Binomial(n, p) when n large via CLT

Modes of Convergence

Definition (Almost Sure Convergence)

A sequence of random variables $(X_n)_{n \geq 1}$ converges to a limit X almost surely (a.s.) or with probability 1 (w.p. 1) if:

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} X_n = X \right) = 1$$

Notation: $X_n \xrightarrow{a.s.} X$ or $X_n \xrightarrow{w.p. 1} X$

Definition (Convergence in Probability)

$(X_n)_{n \geq 1}$ converges in probability to X if for any $\varepsilon > 0$:

$$\lim_{n \rightarrow \infty} \mathbb{P} (|X_n - X| > \varepsilon) = 0$$

Notation: $X_n \xrightarrow{P} X$

Definition (Convergence in Distribution)

Let $(X_n)_{n \geq 1}$ be random variables with CDFs F_n . $(X_n)_{n \geq 1}$ converges in distribution to X (with CDF F) if:

$$\lim_{n \rightarrow \infty} F_n = F$$

Notation: $X_n \xrightarrow{d} X$

Definition (L^r Convergence)

For $r \geq 1$, $(X_n)_{n \geq 1}$ converges in the r -th mean to X if $\mathbb{E}(|X_n|^r), \mathbb{E}(|X|^r)$ exist and:

$$\lim_{n \rightarrow \infty} \mathbb{E}(|X_n - X|^r) = 0$$

Notation: $X_n \xrightarrow{L^r} X$

Remark (Hierarchy of Convergence)

The following implications hold:

$$\text{a.s. conv.} \Rightarrow \text{conv. in } P \Rightarrow \text{conv. in dist.}$$

and

$$L^r \text{ conv.} \Rightarrow \text{conv. in } P \Rightarrow \text{conv. in dist.}$$

Theorem (Strong Law of Large Numbers (SLLN))

Let $(X_n)_{n \geq 1}$ be i.i.d. with $\mu = \mathbb{E}(X_i)$. Then:

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n} \xrightarrow{a.s.} \mu$$

Theorem (Central Limit Theorem (CLT))

Let $(X_n)_{n \geq 1}$ be i.i.d. with $\mu = \mathbb{E}(X_i), \sigma^2 = \text{Var}(X_i) < \infty$. Then:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

The Exponential Distribution

The exponential distribution is fundamental in continuous-time stochastic processes.

Definition (Exponential Distribution)

If $X \sim \text{Exp}(\lambda)$, we have:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Key Properties:

- **CDF:** $F_X(x) = \mathbb{P}(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}$ for $x \geq 0$
- **Tail:** $\mathbb{P}(X > x) = e^{-\lambda x}$
- **Mean:** $\mathbb{E}(X) = \frac{1}{\lambda}$
- **Variance:** $\text{Var}(X) = \frac{1}{\lambda^2}$

Theorem (Memoryless Property)

The exponential distribution is memoryless:

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$$

Proof.

$$\mathbb{P}(X > s + t \mid X > s) = \frac{\mathbb{P}(X > s + t \cap X > s)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(X > t)$$

□

Theorem (Exponential Characterization)

A non-negative random variable X satisfies the memoryless property $\iff$ it has the exponential distribution, i.e., $X \sim \text{Exp}(\lambda)$ for some $\lambda > 0$.

Proposition (Memoryless Property for Random Variables)

If $X \sim \text{Exp}(\lambda)$, then:

$$\mathbb{P}(X > T + s \mid X > T) = \mathbb{P}(X > s)$$

Proposition (Minimum of Exponential Random Variables)

Let $X_1, \dots, X_n$ be independent with $X_i \sim \text{Exp}(\lambda_i)$. Then:

$$\min(X_1, \dots, X_n) \sim \text{Exp}(\lambda_1 + \dots + \lambda_n)$$

and

$$\mathbb{P}\left(X_i = \min_{j=1, \dots, n} X_j\right) = \frac{\lambda_i}{\lambda_1 + \dots + \lambda_n}$$

The Gamma Distribution

The gamma functions extends the domain of the factorial function to positive reals.

Definition (Gamma Function)

For $r > 0$, the gamma function is:

$$\Gamma(r) = \int_0^\infty x^{r-1} e^{-x} dx$$

Properties:

- (a) $\Gamma(r+1) = r\Gamma(r)$
- (b) $\Gamma(1) = 1$
- (c) $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$
- (d) $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Definition (Gamma Distribution)

$X \sim \text{Gamma}(r, \lambda)$ has PDF:

$$f_X(x) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}, \quad x > 0$$

Note: If $r = 1$ and $\lambda > 0$, then $\text{Exp}(\lambda) = \text{Gamma}(1, \lambda)$.

Lemma (Moments of Gamma Distribution)

If $X \sim \text{Gamma}(r, \lambda)$, then:

$$\mathbb{E}(X^n) = \frac{(n+r-1)(n+r-2)\cdots r}{\lambda^n}$$

Key Results:

- For $X \sim \text{Gamma}(r, \lambda)$: $\mathbb{E}(X) = \frac{r}{\lambda}$, $\mathbb{E}(X^2) = \frac{r(r+1)}{\lambda^2}$, $\text{Var}(X) = \frac{r}{\lambda^2}$

Theorem (Sum of i.i.d. Exponentials)

If $X_i \sim \text{Exp}(\lambda)$ i.i.d., then

$$S_n = X_1 + \cdots + X_n \sim \text{Gamma}(n, \lambda)$$

Proposition (Chi-Squared Distribution)

Suppose $Z \sim \mathcal{N}(0, 1)$ and $X = Z^2$. Then:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} x^{-1/2} e^{-x/2}, \quad x > 0$$

This gives us:

$$F_X(x) = \mathbb{P}(Z^2 \leq x) = \mathbb{P}(-\sqrt{x} \leq Z \leq \sqrt{x}) = 2\Phi(\sqrt{x}) - 1$$

where $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-y^2/2} dy$.

The Beta Distribution

Definition (Beta Function)

For $r, s > 0$, the beta function is:

$$B(r, s) = \int_0^1 x^{r-1} (1-x)^{s-1} dx = \frac{\Gamma(r)\Gamma(s)}{\Gamma(r+s)}$$

Definition (Beta Distribution)

$X \sim \text{Beta}(r, s)$ has PDF:

$$f_X(x) = B(r, s)^{-1} x^{r-1} (1-x)^{s-1}, \quad 0 < x < 1$$

Multivariate Normal Distribution

I will not write proofs for this part. Just present results, which we need in Brownian Motion.

Definition (Multivariate Normal Distribution)

Let $\vec{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$ be a sequence of i.i.d. $Z_i \sim \mathcal{N}(0, 1)$. Let A be an $N \times N$ matrix and $\vec{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix}$ be the vector of means. Then $\vec{X}$ is said to have a *multivariate normal distribution* where:

$$X_i = \sum_{k=1}^N a_{ik} Z_k + \mu_i, \quad i = 1, \dots, N$$

Also, a sequence of random variables $X_1, \dots, X_N$ (or vector $\vec{X}$) is said to have MVN distribution if it has the same distribution as $A\vec{Z} + \vec{\mu}$, for some matrix A and sequence of independent standard normals $\vec{Z}$

Lemma

The covariance matrix $\Gamma_{ij} = \text{Cov}(X_i, X_j) = [AA^t]_{ij}$.

Assumption: The covariance matrix is invertible: $\det \Gamma \neq 0$

Theorem (MVN is Characterized by Mean Vector and Covariance Matrix)

Suppose $\vec{X}$ and $\vec{Y}$ have the multivariate normal distribution and equal mean vectors and covariance matrices. That is, suppose $\vec{X}$ has mean vector $\vec{\mu}_1$ and covariance matrix Γ_1 , $\vec{Y}$ has mean vector $\vec{\mu}_2$ and covariance matrix Γ_2 . Then:

$$\vec{\mu}_1 = \vec{\mu}_2 \text{ and } \Gamma_1 = \Gamma_2 \implies \vec{X} \stackrel{d}{=} \vec{Y}$$

Notation: $\text{MVN}(\vec{\mu}, \Gamma)$

Theorem (Multivariate Joint Density)

If $\vec{X} \sim \text{MVN}(\vec{\mu}, \Gamma)$ then the joint density function in vector of the Multivariate Normal Distribution is:

$$f_{\vec{X}}(\vec{x}) = \frac{1}{\sqrt{(2\pi)^N \det \Gamma}} e^{-\frac{1}{2}(\vec{x}-\vec{\mu})\Gamma^{-1}(\vec{x}-\vec{\mu})^T}$$

Theorem (MVN CLT)

If $\vec{X}_1, \dots, \vec{X}_n$ are i.i.d. random vectors with mean vectors $\vec{\mu}$ and covariance matrices Γ then:

$$\lim_{n \rightarrow \infty} \frac{\vec{X}_1 + \dots + \vec{X}_n - n\vec{\mu}}{\sqrt{n}} \stackrel{d}{=} \text{MVN}(\vec{0}, \Gamma)$$

Definition (Powerful MVN Definition)

An equivalent definition of the multivariate normal distribution is as follows:

A random vector $\vec{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$ has the multivariate normal distribution if for any $\vec{a} \in \mathbb{R}^n$ the random variable $a_1 X_1 + \dots + a_n X_n$ has the univariate normal distribution (all linear combinations also normal)

Remark

The above definition is very powerful and often more useful. Also would be quite tough to show, but it is equivalent to the first definition

Note. Note that we will recklessly and with abandon switch between N_t and $N(t)$. They mean the same thing.

Definition (Counting Process)

A continuous-time stochastic process $\{N(t), t \geq 0\}$ is a *counting process* if:

- (a) $N(t) \in \mathbb{N}_0$ (nonnegative integer)
- (b) If $s < t$, then $N(s) \leq N(t)$
- (c) If $s < t$, then $N(t) - N(s) :=$ number of events in the interval $(s, t]$

We define the next two properties in the context of counting process, but they apply to any stochastic process $\{X(t), t \geq 0\}$

Note. The quantity $N(t) - N(s)$ is known as the **increment** and is extremely useful in the study of time-continuous processes.

Definition (Independent Increments)

A counting process $\{N(t), t \geq 0\}$ has *independent increments* if for any $0 = t_0 \leq t_1 < t_2 < \dots < t_n$, the increments $N(t_j) - N(t_{j-1})$ form an independent sequence of RVs for any such sequence:

$$\iff \mathbb{P} \left(\bigcap_{j=1}^n \{N(t_j) - N(t_{j-1}) = k_j\} \right) = \prod_{i=1}^n \mathbb{P}(N(t_j) - N(t_{j-1}) = k_j)$$

Definition (Stationary Increments)

A counting process has *stationary increments* if for any $s \geq 0$ and $0 \leq t_1 < t_2$ the distribution of the increment $N(t_2 + s) - N(t_1 + s)$ does not depend on s , only the length of the interval.

Chapter 2: The Poisson Process

2.1 Introduction

Definition (Poisson Process)

A counting process $(N_t)_{t \geq 0}$ is a Poisson process with rate λ if it satisfies:

1. $N_0 = 0$
2. N_t has independent increments
3. For all $s, t \geq 0$ we have $N_{t+s} - N_s \sim \text{Pois}(\lambda t)$, that is $\mathbb{P}(N_{t+s} - N_s = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$

Remark

The quantity λ is known as the **rate** of the Poisson Process

Note: $\mathbb{E}(N_t) = \lambda t$ (becomes much more important in Chapter 3).

The intuition: Poisson Process models random arrivals and λ is the mean number of arrivals per unit time

Proposition (Poisson Process Joint Distribution)

Let $0 \leq s < t$ and $0 \leq a \leq b$ be integers. Then:

$$\mathbb{P}(N_s = a, N_t = b) = \mathbb{P}(N_s = a) \mathbb{P}(N_t - N_s = b - a) = \frac{e^{-\lambda s} (\lambda s)^a}{a!} \frac{e^{-\lambda(t-s)} (\lambda(t-s))^{b-a}}{(b-a)!}$$

Theorem (Poisson Process Equivalent Characterization)

Let $\{N_t, t \geq 0\}$ be a counting process. Then $\{N_t, t \geq 0\}$ is a Poisson Process with rate $\lambda \iff$

1. $N_0 = 0$
2. The process has stationary increments
3. The process has independent increments
4. $\mathbb{P}(N_h = 1) = \lambda h + o(h)$ and $\mathbb{P}(N_h \geq 2) = o(h)$

Note. We could have equivalently said that if N_t is a counting process with unit jump then N_t is a Poisson Process $\iff N(0) = 0$, N_t has stationary and independent increments

2.2 Interarrival and Arrival Times

Definition (Interarrival Time)

For $n \geq 2$, define $X_n : \Omega \rightarrow \mathbb{R}^+ \cup \{\infty\}$ as the time between the n^{th} event and the $(n-1)^{st}$ event. X_1 is the arrival time of the first event.

Note. In the standard manner for such temporal events, the event

$$\{X_i = \infty\} = \bigcap_{n=1}^{\infty} \{X_i \geq n\}$$

Actually, the fact that there is a first time is something that needs to be established, as there could, in principle, be an infinite number of times arbitrarily close to $t = 0$. However, for a Poisson process:

$$\mathbb{P}(\geq 1 \text{ events in } (0, h]) = \mathbb{P}(N_h \geq 1) = \lambda h + o(h)$$

and

$$\lim_{h \rightarrow 0} \lambda h + o(h) = 0$$

so w.p. 1 there is a first event.

Definition (Arrival Time)

The actual *arrival time* of the n^{th} event is the sum of the first n interarrival times $S_n = X_1 + \dots + X_n$

Theorem

The sequence of interarrival times $(X_n)_{n \geq 1}$ of a Poisson process with rate λ form an i.i.d. sequence of $\text{Exp}(\lambda)$ random variables:

$$X_i \sim \text{Exp}(\lambda)$$

Proof. For the first time we have:

$$\mathbb{P}(X_1 > t) = \mathbb{P}(N_t = 0) = e^{-\lambda t}$$

Now let $n \geq 1$, we have

$$\begin{aligned} \mathbb{P}(X_{n+1} > t) &= \int_0^\infty \mathbb{P}(X_{n+1} > t \mid S_n = s) dF_n(s) = \int_0^\infty \mathbb{P}(N_{s+t} - N_s = 0 \mid S_n = s) dF_n(s) \\ &= \int_0^\infty \mathbb{P}(N_t = 0) dF_n(s) \\ &= e^{-\lambda t} \end{aligned}$$

Independence follows from the memoryless property. □

Corollary (Distribution of S_n)

Since the interarrivals $(X_n)_{n \geq 1}$ are i.i.d. $\text{Exp}(\lambda)$ random variables, $S_n = X_1 + \dots + X_n \sim \text{Gamma}(n, \lambda)$

Proposition (Key Relation)

$$N_t \geq n \iff S_n \leq t$$

The reason is clear: at least n events in $(0, t] \iff n^{\text{th}}$ event happens at t or before. This proposition gives rise to the cdf of the gamma distribution:

$$F_n(t) = \mathbb{P}(S_n \leq t) = \mathbb{P}(N_t \geq n) = \sum_{k=n}^{\infty} \frac{e^{-\lambda t} (\lambda t)^k}{k!}$$

Proposition (Sum of Independent Poisson Processes)

Let $N_1(t), N_2(t)$ be two independent Poisson processes with rate $\lambda_1 + \lambda_2$, then $N_t = N_1(t) + N_2(t)$ is a Poisson Process with rate $\lambda_1 + \lambda_2$

2.3 Conditional Distribution of Arrival Times

Theorem

Given that $N_t = n$, the arrival times $S_1, \dots, S_n$ are distributed as the order statistics of a set of n i.i.d. $\text{Unif}(0, t)$ random variables.

Proof. Use the definition of probability density. □

Chapter 3: Renewal Theory

Renewal process is a generalization from Poisson Process (Poisson Process special case of renewal process), where we again have a counting process N_t representing the number of events in the half-open interval $(0, t]$, but where the interarrivals $X_i \sim F$ for some arbitrary distribution function F .

Definition (Interarrival Times and Distribution)

Let $(X_n)_{n \geq 1}$ be a sequence of i.i.d. random variables with common CDF F . We assume $F(0) = \mathbb{P}(X_n = 0) < 1$, $X_n \geq 0$. The random variable X_n represents the time between the arrival time of the n^{th} event and the $(n-1)$ -st event. Let $\mu = \mathbb{E}(X_i)$, $0 < \mu \leq \infty$

Definition (Arrival Times)

The arrival time of the n^{th} event is $S_n = \sum_{i=1}^n X_i$, $n \geq 1$. Let us also define $S_0 = 0$.

Definition (Renewal Process)

Let $(X_n)_{n \geq 1}$ be interarrival times and $(S_n)_{n \geq 1}$ be the arrival times, as defined above, then the counting process defined by:

$$N_t = \sup\{n : S_n \leq t\}$$

is called a *renewal process*.

Note that by SLLN, $\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu$ a.s., so $\lim_{n \rightarrow \infty} S_n = \infty$ a.s., so w.p. 1 S_n is unbounded.

$$\implies \forall t \geq 0, \exists n > 0 \text{ s.t. } S_n > t \implies S = \{0, 1, \dots, n-1\}$$

is the set of numbers such that $S_n \leq t$ so since $|S| < \infty$, $\sup S = \max S$ (i.e. finite set means max defined).

Remark (Key Relation)

For a renewal process:

$$N_t \geq n \iff S_n \leq t$$

Proof by words: There are n or more arrivals in $(0, t]$ $\iff$ the n^{th} event happened before or at time t

Proposition (S_n Distribution)

The arrival time of the n^{th} event, being the sum of n i.i.d. $X_i \sim F$ has cumulative distribution function $F^{*n}(t)$:

$$F_n(t) := F_{S_n}(t) = F^{*n}(t)$$

where $F^{*n}(t)$ is the distribution of the n -fold convolution of F

Definition (Renewal Function)

A pivotal quantity in renewal theory is the average number of events in $(0, t]$:

$$m(t) = \mathbb{E}(N(t))$$

$m(t)$ is called the **renewal function**

Note. $m(0) = 0$

Theorem (Renewal Function Calculation)

For a renewal process $(N_t)_{t \geq 0}$:

$$m(t) = \sum_{n=1}^{\infty} F_n(t)$$

Proof. We define the indicator function:

$$\mathbb{1}_n = \begin{cases} 1, & \text{if the } n^{th} \text{ arrival is in } (0, t] \\ 0, & \text{else} \end{cases}$$

Then $N_t = \sum_{n=1}^{\infty} \mathbf{1}_n$ so we have

$$\begin{aligned} m(t) = \mathbb{E}(N_t) &= \mathbb{E}\left(\sum_{n=1}^{\infty} \mathbf{1}_n\right) = \sum_{n=1}^{\infty} \mathbb{E}(\mathbf{1}_n) = \sum_{n=1}^{\infty} \mathbb{P}(n^{\text{th}} \text{ arrival in } (0, t]) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(S_n \leq t) \\ &= \sum_{n=1}^{\infty} F_n(t) \end{aligned}$$

□

Proposition (N_t distribution)

$$\mathbb{P}(N_t = n) = F_n(t) - F_{n+1}(t)$$

Proof.

$$\mathbb{P}(N_t = n) = \mathbb{P}(N_t \geq n) - \mathbb{P}(N_t \geq n+1) = \mathbb{P}(S_n \leq t) - \mathbb{P}(S_{n+1} \leq t) = F_n(t) - F_{n+1}(t)$$

□

Remark

The event $\{N_t = n\}$ can be decomposed as follows:

$$\{N_t = n\} = \{S_n \leq t\} \cap \{S_{n+1} > t\}$$

If there are n events by t , the n^{th} arrival must have been by t , and the next one strictly after.

Theorem

For a renewal process $\{N(t), t \geq 0\}$, the expected number of renewals in $(0, t]$ is finite:

$$m(t) < \infty$$

Proof. We begin by noting that since $\mathbb{P}(X_i = 0) < 1$, $\exists a > 0$ such that $\mathbb{P}(X_i \geq a) > 0$

$$\{X_i = 0\} = \bigcap_{n=1}^{\infty} \left\{X_i < \frac{1}{n}\right\}$$

$$\Rightarrow \mathbb{P}(X_i = 0) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} \left\{X_i < \frac{1}{n}\right\}\right) = \lim_{n \rightarrow \infty} \mathbb{P}\left(X_i < \frac{1}{n}\right) < 1 \Leftrightarrow \lim_{n \rightarrow \infty} \mathbb{P}\left(X_i \geq \frac{1}{n}\right) > 0$$

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} \left\{X_i \geq \frac{1}{n}\right\}\right) > 0$$

Define

$$Y_n = \begin{cases} a, & X_n \geq a \\ 0, & X_n < a \end{cases}$$

which is a scaled indicator function: it is a if the n^{th} interarrival is $\geq a$ and 0 otherwise. The sum $\bar{S}_n = \sum_{k=1}^n Y_k$ will count the arrivals of the original process with interarrival $\geq a$ and scale by a .

Define a new renewal process $\bar{N}_t = \sup\{n : \bar{S}_n \leq t\}$. Clearly the renewals happen only at times $na, a \geq 0$. We also define a sequence of stopping times by $T_1 = \min\{k : X_k \geq a\}$, $T_j = \min\{k > T_{j-1} : X_{T_{j-1}+k} \geq a\}$, $j \geq 2$. The random variable T_j represents an interarrival time. Here, time is the index i of X_i and an arrival happens whenever $X_i \geq a$. T_j is the time index of the j^{th} time that $X_i \geq a$. Note that the T_i are i.i.d. To see this we can view the i.i.d. sequence X_i as a Markov chain Z_i which takes value 0, 1 depending on if $X_i \leq$. Thus, since T_i is a stopping time, the process "resets" at each T_j . Thus, i.i.d.

Now, note that $\mathbb{P}(T_1 = k) = \mathbb{P}(X_j < a, 1 \leq j < k, X_k \geq a) \Rightarrow T_j \sim \text{Geom}(\frac{1}{\mathbb{P}(X_m \geq a)})$. Let $p = \frac{1}{\mathbb{P}(X_k \geq a)}$, then $T^{(m)} = T_1 + \dots + T_m$ is the arrival time (in terms of the X_i 's index) of the m^{th} event (when $\mathbb{P}(X_k \geq a)$). Also clearly $T^{(m)} \sim \text{NB}(\frac{1}{p}) \Rightarrow \mathbb{E}(T^{(m)}) = \frac{m}{p}$. Since $T^{(m)}$ is the index of X_i or Y_i we know that $S_{T^{(m)}} = ma$ since the X_k have been greater than or equal to a m times.

Now, it's just floor math and inequalities. Choose $m = \lfloor \frac{t}{a} \rfloor + 1 > \frac{t}{a} \Rightarrow am > t \Rightarrow \bar{S}_{T^{(m)}} > t \Leftrightarrow \bar{N}_t < T^{(m)}$

$$\Rightarrow \mathbb{E}(\bar{N}_t) < \mathbb{E}(T^{(m)}) = \frac{m}{p} = \frac{\lfloor \frac{t}{a} \rfloor + 1}{p}$$

□

Theorem (Renewal Equation)

For a renewal process $\{N(t), t \geq 0\}$, the renewal function can be written as a convolution equation:

$$m(t) = F(t) + \int_0^t m(t-x) dF(x)$$

Proof. This is trivial. Note:

$$\mathbb{E}(N_t | X_1 = s) = \begin{cases} 1 + m(t - s), & \text{if } s \leq t \\ 0, & \text{else} \end{cases}$$

$$\begin{aligned} m(t) &= \mathbb{E}(N_t) \\ &= \mathbb{E}(\mathbb{E}(N_t | X_1)) \\ &= \int_0^\infty \mathbb{E}(N_t | X_1 = s) dF(s) \\ &= \int_0^t (1 + m(t - s)) dF(s) \\ &= \int_0^t dF(s) + \int_0^t m(t - s) dF(s) \\ &= F(t) + \int_0^t m(t - s) dF(s) \end{aligned}$$

□

Remark

The previous equation is $m = F + m \star F$ and the *convolution theorem* says that the Laplace transform of the convolution is the product of the Laplace transforms: $\mathcal{L}(m) = \mathcal{L}(F) + \mathcal{L}(m)\mathcal{L}(F) \implies F = \mathcal{L}^{-1}(\frac{\mathcal{L}(m)}{1+\mathcal{L}(m)})$

Of course the Laplace transform is invertible has some conditions, but in this type of course we assume they hold!

Definition (Poisson Process)

Renewal process with $X_i \sim \text{Exp}(\lambda)$

3.3 - 3.4) Limit Theorems for Renewal Processes

Proposition

Total number of renewals $N_\infty = \lim_{t \rightarrow \infty} N_t = \infty$ w.p. 1

$$\implies \mathbb{P}(N_\infty < \infty) = \mathbb{P}(\text{finite \# of renewals})$$

$$\begin{aligned}
&= \mathbb{P}(\exists n \geq 0 \text{ s.t. } X_n = \infty) \\
&= \mathbb{P}\left(\bigcup_{n=1}^{\infty} \{X_n = \infty\}\right) \\
&\leq \sum_{n=1}^{\infty} \mathbb{P}(X_n = \infty) \\
&= \sum_{n=1}^{\infty} \lim_{t \rightarrow \infty} \mathbb{P}(X_n > t) \\
&= \sum_{n=1}^{\infty} 0 = 0
\end{aligned}$$

Theorem (SLLN for Renewal Processes)

$$\lim_{t \rightarrow \infty} \frac{N_t}{t} = \frac{1}{\mu}$$

Proof. Note that $S_{N_t} \leq t < S_{N_t+1} \implies \frac{S_{N(t)}}{N(t)} \leq \frac{t}{N(t)} < \frac{S_{N(t)+1}}{N(t)} \frac{N(t)+1}{N(t)+1}$. Note that since $N_t \rightarrow \infty, t \rightarrow \infty$, we have by SLLN

$$\begin{aligned}
\lim_{t \rightarrow \infty} \frac{S_{N(t)}}{N(t)} &\leq \lim_{t \rightarrow \infty} \frac{N_t}{t} < \lim_{t \rightarrow \infty} \frac{S_{N(t)+1}}{N(t)+1} \lim_{t \rightarrow \infty} \frac{N(t)+1}{N_t} \\
\mu &\leq \lim_{t \rightarrow \infty} \frac{N_t}{t} < \mu * 1
\end{aligned}$$

Squeeze theorem $\implies$ result □

Wald's Equation

Let $X_1, X_2, \dots$ denote a sequence of independent random variables with $\mu = \mathbb{E}(X_i)$

Definition (Stopping Time)

A nonnegative integer valued random variable N is a stopping time for $(X_n)_{n \geq 1}$ if the event $\{N = n\}$ is independent of $(X_k)_{k > n} \Leftrightarrow \{N = n\}$ is fully determined by $X_1, \dots, X_n \Leftrightarrow \mathbf{1}_{\{N=n\}} = f(X_1, \dots, X_n)$

Theorem (Wald's Equations)

If $\mu < \infty$ and N is a stopping time for $(X_n)_{n \geq 1}$ with $\mathbb{E}(N) < \infty$ then we have $\mathbb{E}(S_N) = \mathbb{E}(N) \mathbb{E}(X_1)$

Proof. Indicators! Define $I_n = \begin{cases} 1 & n \leq N \\ 0 & n > N \end{cases}$
 $\Rightarrow S_N = \sum_{i=1}^N X_i = \sum_{n=1}^{\infty} X_n I_n$

$$\Rightarrow \mathbb{E}(S_N) = \mathbb{E}\left(\sum_{n=1}^{\infty} X_n I_n\right) = \sum_{n=1}^{\infty} \mathbb{E}(X_n I_n) = \sum_{n=1}^{\infty} \mathbb{E}(X_n) \mathbb{E}(I_n) \quad (3)$$

$$= \sum_{n=1}^{\infty} \mu \mathbb{P}(N \geq n) = \mu \mathbb{E}(N) \quad (4)$$

$X_n \perp I_n$ because $I_n = \mathbf{1}_{\{N \geq n\}} \Leftrightarrow \{N \geq n\}^c = \{N \leq n-1\}$, $\mathbf{1}_{\{N \leq n-1\}} = g(X_1, \dots, X_{n-1})$
 $\square \qquad \qquad \qquad \square$

Elementary Renewal Theorem

Theorem (Elementary Renewal Theorem)

Let $m(t) = \mathbb{E}(N_t)$, $\mu = \mathbb{E}(X_i)$, then $\lim_{t \rightarrow \infty} \frac{m(t)}{t} = \frac{1}{\mu}$ (ERT)

Proof. First, we claim that $N_t + 1$ is a stopping time for $(N_t)_{t \geq 0}$. This follows from the fact that

$$\{N_t + 1 \leq n\} = \{N_t \leq n-1\} = \{S_{n-1} \leq t\}$$

clearly determined by $X_1, \dots, X_n$. Thus, by Wald's Equation

$$\Rightarrow \mathbb{E}(S_{N_t+1}) = \mu \mathbb{E}(N_t + 1) = \mu[m(t) + 1]$$

Obviously, $S_{N_t+1} > t \Rightarrow \mu(m(t) + 1) > t \Rightarrow \frac{m(t)}{t} > \frac{1}{\mu} - \frac{1}{t}$

Recall that for a function $f(t)$,

$$\liminf_{t \rightarrow \infty} f(t) = \lim_{t \rightarrow \infty} \inf_{\gamma \geq t} f(\gamma)$$

, so let $\gamma \geq t \Rightarrow \frac{1}{\gamma} \leq \frac{1}{t} \Rightarrow -\frac{1}{\gamma} \geq -\frac{1}{t}$

So, we have $\frac{m(\gamma)}{\gamma} \geq \frac{1}{\mu} - \frac{1}{\gamma} \geq \frac{1}{\mu} - \frac{1}{t} \Rightarrow \inf_{\gamma \geq t} \frac{m(\gamma)}{\gamma} \geq \frac{1}{\mu} - \frac{1}{t} \Rightarrow \liminf_{t \rightarrow \infty} \frac{m(t)}{t} \geq \frac{1}{\mu}$

Now, let's fix $M > 0$ and define new renewal process

$$\bar{X}_n = \begin{cases} X_n & \text{if } X_n \leq M \\ M & \text{if } X_n > M \end{cases}$$

so we have the analogous arrival times and counting functions

$$\bar{S}_n = \sum_{i=1}^n \bar{X}_i, \bar{N}_t = \sup\{n : \bar{S}_n \leq t\}$$

Note that $\bar{S}_{\bar{N}_t} \leq t \implies \bar{S}_{\bar{N}_t} + \bar{X}_{\bar{N}_t+1} \leq t + M \implies \bar{S}_{\bar{N}_t+1} \leq t + M$

$$\begin{aligned} \mu_M(\bar{m}(t) + 1) \leq t + M &\Leftrightarrow \frac{\bar{m}(t)+1}{t} \leq \frac{1+\frac{M}{t}}{\mu_M} \\ \Rightarrow \limsup_{t \rightarrow \infty} \frac{\bar{m}(t)}{t} &= \lim_{t \rightarrow \infty} \sup_{\gamma \geq t} \frac{\bar{m}(\gamma)}{\gamma}, \text{ for } \gamma \geq t \text{ we have} \end{aligned}$$

$$\frac{\bar{m}(\gamma)}{\gamma} \leq \frac{1}{\mu_M} \left(1 + \frac{M}{\gamma}\right) - \frac{1}{\gamma} < \frac{1}{\mu_M} + \frac{1}{\gamma} \leq \frac{1}{\mu_M}$$

Thus "least upper bound" for $\frac{\bar{m}(\gamma)}{\gamma}$: $\sup_{\gamma \geq t} \frac{\bar{m}(\gamma)}{\gamma} \leq \frac{1}{\mu_M} \Rightarrow \limsup_{t \rightarrow \infty} \frac{\bar{m}(t)}{t} \leq \frac{1}{\mu_M}$

Since $\bar{S}_n \leq S_n$ we have $\bar{N}(t) \geq N(t)$ so $\bar{m}(t) \geq m(t)$, thus $\limsup_{t \rightarrow \infty} \frac{m(t)}{t} \leq \frac{1}{\mu_M}$

$$\lim_{M \rightarrow \infty} \mu_M = \mu \Rightarrow \limsup_{t \rightarrow \infty} \frac{m(t)}{t} \leq \frac{1}{\mu}$$

$$\limsup_{t \rightarrow \infty} \frac{m(t)}{t} \leq \frac{1}{\mu} \text{ and } \liminf_{t \rightarrow \infty} \frac{m(t)}{t} \geq \frac{1}{\mu} \Rightarrow \lim_{t \rightarrow \infty} \frac{m(t)}{t} = \frac{1}{\mu} \quad \square$$

Theorem (Renewal CLT)

Let $\mu = \mathbb{E}(X_i)$, $\sigma^2 = \text{Var}(X_i)$ with both $\mu, \sigma < \infty$. Then:

$$\lim_{t \rightarrow \infty} \mathbb{P} \left(\frac{N_t - \frac{t}{\mu}}{\sigma \sqrt{\frac{t}{\mu^3}}} < y \right) = \Phi(y)$$

Proof Omitted. Uninstructive

Blackwell's Theorem

Definition (Lattice Random Variable)

A nonnegative random variable X is lattice if there exists $d \geq 0$ such that $\sum_{n=0}^{\infty} \mathbb{P}(X = nd) = 1$. In other words, a lattice RV is one that can only take on discretized values (integer multiples of d). The largest such d is known as the *period* of the lattice RV.

Remark (Lattice RV Special Case)

A discrete random variable is a lattice random variable with $d = 1$

Theorem (Blackwell's Theorem)

Let $(N_t)_{t \geq 0}$ be a renewal process with interarrivals $(X_i)_{i \geq 1}$ such that $X_i \sim F$ and $\mu = \mathbb{E}(X_i)$:

1. If F is not lattice, then $\lim_{t \rightarrow \infty} [m(t+a) - m(t)] = \frac{a}{\mu}$
2. If F is lattice with period d , then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(\text{number of renewals at } nd) = \lim_{n \rightarrow \infty} [m(nd) - m((n-1)d)] = \frac{d}{\mu}$$

Note. Proof omitted.

If simultaneous arrivals are allowed $\mathbb{P}(X_i > 0) > 0$, then in the lattice case the jump might be more than one, but if not it is restricted to unit jump

Remark (DRI Sufficient Condition)

If $h : [0, \infty) \rightarrow \mathbb{R}$ satisfies:

1. $h(t) \geq 0, t \geq 0$
2. $h(t)$ is nonincreasing
3. $\int_0^\infty h(t)dt < \infty$

Theorem (Key Renewal Theorem)

If F is not lattice and if $h(t)$ is Directly Riemman Integrable (DRI):

$$\lim_{t \rightarrow \infty} \int_0^t h(t-x)dm(x) = \frac{1}{\mu} \int_0^\infty h(t)dt$$

Note. We will not prove this, but instead move on to more interesting topics!

Theorem (CDF Last Arrival Time)

Obviously S_{N_t} is the last arrival time, then:

$$F_{S_{N(t)}}(s) = \mathbb{P}(S_{N(t)} \leq s) = \bar{F}(t) + \int_0^s \bar{F}(t-y)dm(y), 0 \leq s \leq t$$

Proof.

$$\begin{aligned}
\mathbb{P}(S_{N_t} \leq s) &= \sum_{n=0}^{\infty} \mathbb{P}(S_{N_t} \leq s, N_t = n) \\
&= \sum_{n=0}^{\infty} \mathbb{P}(S_n \leq s, S_n \leq t, S_{n+1} > t) \\
&= \mathbb{P}(X_1 > t) + \sum_{n=1}^{\infty} \int_0^s \mathbb{P}(S_{n+1} > t) dF_n(y) \\
&= \bar{F}(t) + \sum_{n=1}^{\infty} \int_0^s \mathbb{P}(S_n + X_{n+1} > t) dF_n(y) \\
&= \bar{F}(t) + \int_0^s \bar{F}(t-y) d\left(\sum_{n=1}^{\infty} F_n(y)\right) \\
&= \bar{F}(t) + \int_0^s \bar{F}(t-y) dm(y)
\end{aligned}$$

$$\Rightarrow F_{S_{N_t}}(s) = \mathbb{P}(S_{N_t} \leq s) \Rightarrow \frac{dF_{S_{N_t}}(s)}{ds} = \bar{F}(t-s)m'(s)$$

□

Corollary

For $s \leq t$

$$dF_{S_{N(t)}}(s) = \bar{F}(t-s)dm(s)$$

Alternating Renewal Processes

We begin with a little talk. The scenario is that we have 2-state system, where the state is either *on* or *off*. The system starts on and remains on for time Z_1 . After time Z_1 , the system switches off and remains off for time Y_1 . Then it cycles back on for time Z_2 , off for time Y_2 , and so on... You get the picture.

Assumption: The random vectors $(Z_n, Y_n), n \geq 1$ are i.i.d. but crucially Z_n and Y_n might be dependent.

Conceptually this partitions time into i.i.d. cycles on (on, off). But inside a cycle, the on period can depend on the off period, but overall the cycles are i.i.d. Also the length of time between successive arrivals of the "on" state is $Z_n + Y_n$

Notation. In terms of their CDFs, $Z_n \sim H, Y_n \sim F, Z_n + Y_n \sim G, n \geq 1$. Also define

$$P(t) = \mathbb{P}(\text{system on at time } t)$$

Definition

Define the state space $\mathcal{S} = \{0, 1\}$ and $\{X(t), t \geq 0\}$ represents the state of the system as in $X(t) \in \mathcal{S}$. Let Z_n be the amount contiguous of time $X(t) = 1$ before it switches to 0 and let Y_n be the amount of contiguous time $Y(t) = 0$ before it switches to 1.

Theorem

If $\mathbb{E}(Z_n + Y_n) < \infty$ and F is nonlattice, then:

$$\lim_{t \rightarrow \infty} P(t) = \frac{\mathbb{E}(Z_n)}{\mathbb{E}(Z_n + Y_n)}$$

Proof. This is a very important proof. Let us define a renewal process $N(t)$ that counts the number of times the system switches on in $(0, t]$. The interarrival distribution is $Z_n + Y_n \sim F$. We will compute $P(t)$ by conditioning on $S_{N(t)}$, which is the number of renewals by time t . Note that:

$$\begin{aligned} P(t) &= \mathbb{P}(X(t) = 1) \\ &= \mathbb{P}(X(t) = 1 \mid S_{N(t)} = 0) \mathbb{P}(S_{N(t)} = 0) + \int_0^t \mathbb{P}(X(t) = 1 \mid S_{N_t} = y) dF_{S_{N(t)}}(y) \end{aligned}$$

Now, let's compute the first term:

$$\begin{aligned} \mathbb{P}(X(t) = 1 \mid S_{N(t)} = 0) &= \mathbb{P}(Z_1 > t \mid Z_1 + Y_1 > t) \\ &= \frac{\mathbb{P}(Z_1 > t \cap Z_1 > t - Y_1)}{\mathbb{P}(Z_1 + Y_1 > t)} \\ &= \frac{\mathbb{P}(Z_1 > t)}{\mathbb{P}(Z_1 + Y_1 > t)} \\ &= \frac{\bar{H}(t)}{\bar{F}(t)} \end{aligned}$$

Now, for the second term note that conditional on $S_{N(t)} = y$, we can reason as follows. There have been y renewals by time t , the (on, off) cycles being i.i.d, thus what we want is for the current "on" period $Z > t - y$. Also, we want the length of the current cycle to be greater than the elapsed time of $t - y$. Thus:

$$\begin{aligned} \mathbb{P}(X_t = 1 \mid S_{N(t)} = y) &= \mathbb{P}(Z > t - y \mid Z + Y > t - y) \\ &= \frac{\mathbb{P}(Z > t - y \cap Z > t - y - Y)}{\mathbb{P}(Z + Y > t - y)} \\ &= \frac{\bar{H}(t - y)}{\bar{F}(t - y)} \end{aligned}$$

Now, plugging things back in, we have:

$$\begin{aligned} P(t) &= \mathbb{P}(X(t) = 1 \mid S_{N(t)} = 0) \mathbb{P}(S_{N(t)} = 0) + \int_0^\infty \mathbb{P}(X(t) = 1 \mid S_{N_t} = y) dF_{S_{N(t)}}(y) \\ &= \frac{\bar{H}(t)}{\bar{F}(t)} \bar{F}(t) + \int_0^t \frac{\bar{H}(t-y)}{\bar{F}(t-y)} dF_{S_{N(t)}}(y) \end{aligned}$$

Note that $F_{S_{N(t)}}(s) = \mathbb{P}(S_{N(t)} \leq s) = \bar{F}(t) + \int_0^s \bar{F}(t-y) dm(y)$, $0 \leq s \leq t$ so that $\frac{dF_{S_{N(t)}}(s)}{ds} = \bar{F}(t-s)m'(s) \implies dF_{S_{N(t)}}(s) = \bar{F}(t-s)dm(s)$. So, plugging into the formula for $P(t)$ yields:

$$P(t) = \bar{H}(t) + \int_0^t \bar{H}(t-y) dm(y)$$

We can apply the Key Renewal Theorem on $h(t) = \bar{H}(t)$ (clearly DRI, as it is a CDF), so we get

$$\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow 0} \bar{H}(t) + \lim_{t \rightarrow \infty} \int_0^t \bar{H}(t-y) dm(y) = \frac{1}{\mathbb{E}(Z_n + Y_n)} \int_0^\infty \bar{H}(t) dt = \frac{\mathbb{E}(Z_n)}{\mathbb{E}(Z_n + Y_n)}$$

□

3.5 Delayed Renewal Processes

Definition

A delayed renewal process $N_D(t)$ is the same in all aspects except that the first inter-arrival time has a different distribution G . The rest of them have distribution F .

Remark

Not much more interesting theory to say, but obviously all limit laws hold from renewal theory.

We can use renewal theory to solve interesting problems. The following problem can be solved in this way as a consequence of Blackwell's

3.5.1) Pattern Problems

Scenario: Toss coins X_i . $\mathbb{P}(X_i = H) = p, q = 1 - p = \mathbb{P}(X_i = T)$

Remark: The expected number of tosses between occurrences (pattern arrival is when the last letter in the pattern appears) is $\frac{1}{p^k q^j}$ for k heads j tails.

Formula: Analyze the pattern P . If it begins and ends the same way (some prefix is the same as some suffix), then

$$\mathbb{E}(\text{tosses to } P) = \mathbb{E}(\text{tosses to prefix}) + \frac{1}{p^k q^j}$$

Start with the longest prefix and then keep decomposing further until you get a simple pattern with no such (prefix = suffix) property

Proposition (Pattern Computation Formulas)

Definitions:

$N_A := \#$ of tosses to pattern A

$N_{B|A} := \#$ of additional flips needed to get B , starting with A

$P_A := \mathbb{P}(A \text{ before } B)$

$M = \min\{N_A, N_B\}$ (Note $P_A = \mathbb{P}(M = A)$)

Derivation:

$$\begin{aligned} \Rightarrow \mathbb{E}(N_A) &= \mathbb{E}(M) + \mathbb{E}(N_A - M) \\ &= \mathbb{E}(M) + \mathbb{E}(\mathbb{E}(N_A - M \mid M)) \\ &= \mathbb{E}(M) + \mathbb{E}(N_{A|A}) P_A + \mathbb{E}(N_{A|B}) [1 - P_A] \\ &= \mathbb{E}(M) + \mathbb{E}(N_{A|B}) [1 - P_A] \end{aligned}$$

$$\Rightarrow \mathbb{E}(N_B) = \mathbb{E}(M) + \mathbb{E}(N_{B|A}) P_A$$

Solve to get

Final Formulas:

$$P_A = \frac{\mathbb{E}(N_B) + \mathbb{E}(N_{A|B}) - \mathbb{E}(N_A)}{\mathbb{E}(N_{B|A}) + \mathbb{E}(N_{A|B})}$$

$$\mathbb{E}(M) = \mathbb{E}(N_B) - \mathbb{E}(N_{B|A}) P_A$$

Example: On the midterm, we were asked to find the mean number of flips until the pattern

$$P = THTHTHTHT$$

first occurs.

$$\mathbb{E}(N_P) = \mathbb{E}(N_{THTHT}) + \frac{1}{p^4 q^5}$$

$$\begin{aligned}
&= \mathbb{E}(N_{THT}) + \frac{1}{p^2 q^3} + \frac{1}{p^4 q^3} \\
&= \mathbb{E}(N_T) + \frac{1}{pq^2} + \frac{1}{p^2 q^3} + \frac{1}{p^4 q^3} \\
&= \frac{1}{q} + \frac{1}{pq^2} + \frac{1}{p^2 q^3} + \frac{1}{p^4 q^3}
\end{aligned}$$

3.6 Renewal Reward

We have the following setup for a renewal-reward process, which is a slight variation. We have a renewal process N_t , as before, but each time a renewal occurs we get a reward. Denote by R_n the reward earned from the n^{th} renewal. Note that $R_n = f(X_n)$ and we assume that the pairs (X_n, R_n) , $n \geq 1$ are independent and identically distributed as well as the individual sequences $(X_n)_{n \geq 1}$, $(R_n)_{n \geq 1}$.

Definition (Total Reward)

The total reward earned by a Renewal-Reward Process is given by $R(t) = \sum_{n=1}^{N(t)} R_n$

Theorem (Renewal-Reward Theorem)

If $\mathbb{E}(R_n) < \infty$ and $\mathbb{E}(X_n) < \infty$, then:

1. with probability 1, $\frac{R(t)}{t} \rightarrow \frac{\mathbb{E}(R_n)}{\mathbb{E}(X_n)}$, $t \rightarrow \infty$
2. $\frac{\mathbb{E}(R(t))}{t} \rightarrow \frac{\mathbb{E}(R_n)}{\mathbb{E}(X_n)}$, $t \rightarrow \infty$

Proof. (i) We have

$$\frac{R(t)}{t} = \frac{\sum_{i=1}^{N(t)} R_i}{t} = \frac{\sum_{i=1}^{N(t)} R_i}{N(t)} \frac{N(t)}{t}$$

By SLLN (general and for renewals), we have $\lim_{t \rightarrow \infty} \frac{R(t)}{t} = \mathbb{E}(R_n) \frac{1}{\mathbb{E}(X_i)}$

(ii) Too complicated, hence omitted. See Ross. □

Regenerative process, Random Walk and arcsine

Consider a stochastic process $\{X(t), t \geq 0\}$ with state space $\mathcal{S} = \{0, 1, 2, \dots\}$ that has the property that with probability 1, there exist times $T_1, T_2, \dots$ at which the process resets itself. From times T_i onward the process has the same distribution as the process starting from 0.

We can define a renewal process $N(t)$ which counts the number of such regeneration points. Let us say a cycle is completed everytime the process resets itself.

Definition (Regenerative Process)

A stochastic process $\{X(t), t \geq 0\}$ for which there exist a sequence of times $(S_i)_{i \geq 1}$ such that the process restarts itself probabilistically at each time T_i :

$$X(S_i + t) \stackrel{d}{=} X(t)$$

This means that the process resets to 0 at each time S_i . The times S_i are the event times of the renewal process

$$N_S(t) := \text{number of resets by in } (0, t]$$

A **cycle** occurs each time a renewal occurs. We have $X_i \sim F$ as the interarrival times representing cycle times.

The following theorem states that:

$$P_j = \lim_{t \rightarrow \infty} \mathbb{P}(X(t) = j) = \frac{\mathbb{E}(\text{amount of time in state } j \text{ during a cycle})}{\mathbb{E}(\text{cycle length})}$$

Proposition

The time spent in a cycle has distribution S_1 . Thus $X_i \sim S_1 \sim F$

No proof required, this is simply because the process resets itself at each time S_i and the cycles, being interarrivals, are i.i.d.

Proposition

The distribution of the n^{th} arrival time is the same as the 1^{st} :

$$F_{S_n}(t) = F_{S_1}(t)$$

Again, no proof required, but this is tricky to see, so I explain! In an ordinary renewal process and in this one, S_n is the n^{th} arrival time, the time between 0 and that of arrival n . The interarrival $X_n = S_n - S_{n-1} \sim S_1 = X_1$ again this is true for any renewal process, but crucial in this is the fact that starting from any regeneration looks like starting from 0, probabilistically. Thus $S_n \sim S_1$

Theorem

If F has a density over some interval I and $\mathbb{E}(S_1) < \infty$, then:

$$P_j = \lim_{t \rightarrow \infty} \mathbb{P}(X(t) = j) = \frac{\mathbb{E}(T_j)}{\mathbb{E}(S_1)}$$

where $T_1 = \int_0^{S_1} \mathbb{1}_{\{X(t)=j\}} dt$ represents the time spent in state j during a cycle (all have length i.i.d. to S_1)

Proof. Similarly to the alternating renewal theorem, we can condition on the last renewal (regeneration):

$$\begin{aligned} P(t) &= \mathbb{P}(X(t) = j) = \mathbb{P}(X(t) = j \mid S_{N(t)} = 0) \mathbb{P}(S_{N(t)} = 0) \\ &\quad + \int_0^t \mathbb{P}(X(t) = j \mid S_{N(t)} = s) dF_{S_{N(t)}}(s) \\ &= \mathbb{P}(X(t) = j \mid S_1 > t) \mathbb{P}(S_1 > t) \\ &\quad + \int_0^t \mathbb{P}(X(t-s) = j \mid S_1 > t-s) \bar{F}(t-s) dm(s) \end{aligned}$$

Note that for $\mathbb{P}(X(t) = j \mid S_{N(t)} = s)$, we condition on last renewal occurring at time s so time in current cycle is $t-s$. Since probabilistically, process restarts at s , we can shift to $X(t-s)$ and the condition is that the current cycle's length (i.i.d to S_1) is greater than $t-s$. In general, we proved that for a renewal process $dF_{S_{N(t)}}(s) = \bar{F}(t-s) dm(s)$ where F interarrival distribution (here $S_1 \sim F$ is inter-arrival time)

$$\implies P(t) = \mathbb{P}(X(t) = j, S_1 > t) + \int_0^t \mathbb{P}(X(t-s) = j, S_1 > t-s) dm(s)$$

□

Let $h(t) = \mathbb{P}(X(t) = j, S_1 > t)$, clearly DRI, then apply KRT:

$$\implies \lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} h(t) + \lim_{t \rightarrow \infty} \int_0^t h(t-s) dm(s) = \frac{1}{\mathbb{E}(S_1)} \int_0^\infty \mathbb{P}(X(t) = j, S_1 > t) dt$$

(survival tail is 0) remains to show the numerator is what we want, but this is easy:

$$\mathbb{E}(T_1) = \mathbb{E}\left(\int_0^{S_1} \mathbb{1}_{\{X(t)=j\}} dt\right) = \int_0^\infty \mathbb{E}(\mathbb{1}_{\{X(t)=j, t < S_1\}}) dt = \int_0^\infty \mathbb{P}(X(t) = j, t < S_1) dt$$

Note. This theorem establishes the limiting probability distribution α_j , while the next one concerns the long-run proportion of time spent in a state

$$A_j(t) := \text{amount of time in state } j \text{ during } (0, t) = \int_0^t \mathbb{1}_{\{X(s)=j\}} ds$$

Proposition (Long-Run Regenerative Proportions)

For a regenerative process with $\mathbb{E}(S_1) < \infty$, with probability 1,

$$\lim_{t \rightarrow \infty} \frac{A_j(t)}{t} = \frac{\mathbb{E}(T_1)}{\mathbb{E}(S_1)}$$

Proof. We want to try to fit a renewal-reward framework. The cycles/interarrivals distributed $X_n \sim S_1$ are the interarrivals and we define a reward $R_i(X_i) = \int_0^{X_i} \mathbb{1}_{\{X(t)=j\}} \leftarrow$ amount of time in j during a cycle, thus what we want is the amount of time in j up to time t , $\implies R(t) = \sum_{i=1}^{N(t)} R_i(X_i)$ (add up all the cycles) By ALT:

$$\frac{R_t}{t} \rightarrow \frac{\mathbb{E}(R_i)}{\mathbb{E}(X_i)}$$

□

This establishes the long-run proportions $= P_j$.

Chapter 4: Markov Chain

This whole section was the entire last quarter! I will just state the things important in CTMC, Martingale, Brownian Motion.

Markov Chains and Markov Properties

Definition (Markov Chain - Discrete Time)

A sequence of random variables $\{X_n\}_{n \geq 0}$ on a countable state space $\mathcal{S}$ is a *Markov chain* if it satisfies the **Markov property**:

For all $n \geq 0$ and states $i, j, i_0, i_1, \dots, i_{n-1} \in \mathcal{S}$:

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

Intuitive meaning: Given the present state, the future is independent of the past.

Definition (Stopping Time)

A random variable $T : \Omega \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$ is a *stopping time* with respect to the filtration $(\mathcal{F}_n)$ if:

$$\{T = n\} \in \mathcal{F}_n \text{ for all } n \geq 0$$

Intuitive meaning: The decision to stop at time n can be made based only on information available up to time n .

Theorem (Strong Markov Property - Discrete Time)

A Markov Chain restarts at a stopping time T . We can thus say that:
 $\mathbb{P}(X_{T+1} = j \mid X_T = i, X_k, 0 \leq k < T) = \mathbb{P}(X_{T+1} = j \mid X_T = i)$

Remark (Key Distinction)

- **Markov Property:** Applies at fixed, deterministic times
- **Strong Markov Property:** Applies at random stopping times
- The strong Markov property is crucial for analyzing hitting times, absorption probabilities, and other random time events
- Not all processes with the Markov property have the strong Markov property, but most "nice" Markov processes do

Gambler's Ruin Problem

Definition (Setup)

Consider a random walk on states $\{0, 1, 2, \dots, b\}$ where:

- Start with wealth x where $a < x < b$
- Move right (+1) with probability p , left (-1) with probability $q = 1 - p$
- States $a = 0$ and b are absorbing (ruin and target wealth)
- $T = T^{\{a,b\}} = \min\{n \geq 0 : X_n \in \{a, b\}\}$ is the absorption time

Theorem (Gambler's Ruin Formulas)

Let $u_x = \mathbb{P}(T^a < T^b \mid X_0 = x)$ be the ruin probability starting from x .

Ruin Probability:

$$u_x = \begin{cases} \frac{b-x}{b-a} & \text{if } p = q = \frac{1}{2} \\ \frac{\left(\frac{q}{p}\right)^{x-a} - \left(\frac{q}{p}\right)^{b-a}}{1 - \left(\frac{q}{p}\right)^{b-a}} & \text{if } p \neq q \\ 1 & \text{if } x = a \\ 0 & \text{if } x = b \end{cases}$$

Chapter 5: Continuous Time Markov Chains

Continuous time Markov chain (CTMCs) are extensions of discrete-time chains. We may also think of it as a Poisson process without unit jump.

Definition (Continuous-Time Markov Chain)

A stochastic process $\{X(t)\}_{t \geq 0}$ on a countable state space $\mathcal{S}$ (i.e. $|\mathcal{S}| < \infty$) is a *Continuous Time Markov Chain* (CTMC) if for all $s, t \geq 0$ and states $i, j \in \mathcal{S}$,

$$\mathbb{P}(X(t+s) = j \mid X(s) = i, X(u) = s_u, 0 \leq u < s) = \mathbb{P}(X(t+s) = j \mid X(s) = i).$$

This is known as the **Markov property** (in continuous time).

Definition (Stationary/Time-homogeneous Process)

The CTMC $(X(t))_{t \geq 0}$ is stationary or time-homogeneous if $\mathbb{P}(X(t+s) = j \mid X(s) = i)$ is independent of s .

Intuition: Time shift by s doesn't affect the distribution.

From now on, we assume stationarity, and thus we define the transition function $P_{ij}(t) = \mathbb{P}(X(t+s) = j \mid X(s) = i)$.

Definition (Transition Function)

We define the transition function as:

$$P_{ij}(t) = \mathbb{P}(X(s+t) = j \mid X(s) = i)$$

Note that this is a time dependent matrix!

Proposition (Transition Function Stochastic)

Fix $t \geq 0$. Then:

$$\sum_{j \in \mathcal{S}} P_{ij}(t) = 1$$

that is, for a fixed time $P_{ij}(t)$ is a stochastic matrix

Informally the **holding time** τ_i of state $i \in \mathcal{S}$ is the amount of time the process spends in state i after it enters and before it jumps to another state.

Note (CTMC Strong Markov Property). Let T be a stopping time (i.e. the event $\{T \leq t\}$ can be determined by $X(s), 0 \leq s \leq t$), then Markov property holds for T too.

$$\mathbb{P}(X(T+t) = j \mid X(T) = i, X(u), 0 \leq u < T) = \mathbb{P}(X(T+t) = j \mid X(T) = i) = P_{ij}(t).$$

Proof. Beyond the scope of the course. □

Definition (Hitting Times)

Let $i \in \mathcal{S}$. Define the sequence $T_n^i, n \geq 1$ by $T^i = T_1^i = \inf\{t > 0 : X(t) = i\}$ and $T_n^i = \inf\{t > T_{n-1}^i : X(t) = i\}, n > 1$.

Definition (Holding Times)

Let T be a hitting time of x , then:

$$\tau_x := \inf\{t > 0 : X(T+t) \neq x\}$$

Strong Markov means that this has the same distribution regardless of which hitting time it is, since the process resets at each hitting time, allowing this definition to be well founded!

Proposition (τ_i distribution)

The holding time τ_i associated with state i has exponential distribution: $\tau_i \sim \text{Exp}(\nu_i)$ where the parameter ν_i is the rate at which the process transitions out of state i .

Proof.

$$\begin{aligned}
\mathbb{P}(\tau_i > s + t \mid \tau_i > s) &= \mathbb{P}(X(u) = i, T \leq u \leq T + s + t \mid X(u) = i, T \leq u \leq T + s) \\
&= \mathbb{P}(X(u) = i, T + s < u \leq T + s + t \mid X(T + s) = i) \\
&= \mathbb{P}(X(u) = i, T < u \leq T + t \mid X(T) = i) \quad (\text{stationary}) \\
&= \mathbb{P}(\tau_i > t)
\end{aligned}$$

□

Let us discuss this result intuitively. Suppose a Continuous-Time Markov chain enters state i at some time and stays in that state for an amount of time bigger than s . Then the state $X(t) = i$ for s time units since it entered. We can "get rid" of the past, so the probability that it stays for another t , does not depend on what happened during last s time units, and since the current state is i , we have basically reset the process and have the same situation as $\tau_i > t$.

Also the exponential distribution's memoryless property in continuous time roughly translates to the Markov property. The Poisson process is a special case where we are restricted to the unit jump (i.e. can only transition to 1 up).

Proposition

τ_i is independent of the next state

Proof. Follows easily from Markov property. Suppose currently $X(0) = i$ and state i has holding time $\tau_i \sim \text{Exp}(\nu_i)$. Let S_i be the next state given the current state is $i \in \mathcal{S}$. Then: $\mathbb{P}(S_i \mid \tau_i > s) = \mathbb{P}(S_i \mid X(s) = i, 0 \leq t \leq s) = \mathbb{P}(S_i \mid X(s) = i) = \mathbb{P}(S_i)$ □

Definition (Regularity)

A CTMC is regular $\iff$ the number of transitions in a time interval of finite length is finite with probability 1

Remark (Standing Regularity Assumption)

From now on, assume all CTMC we discuss are regular.

Note. $\mathbb{P}(\tau_i = \infty) = \lim_{t \rightarrow \infty} \mathbb{P}(\tau_i > t) = \lim_{t \rightarrow \infty} e^{-\nu_i t}$ and $\mathbb{P}(\tau_i = 0) = \lim_{t \rightarrow 0} \mathbb{P}(\tau_i < t) = 1 - \lim_{t \rightarrow \infty} e^{-\nu_i t}$

Definition (Absorbing State)

A state $i \in \mathcal{S}$ is absorbing if $\nu_i = 0$

This means the mean time spent in state i is $\frac{1}{0} = \infty$.

Definition (Instantaneous State)

A state $i \in \mathcal{S}$ is instantaneous if $\nu_i = \infty$,

This means that the mean time spent in state i is 0, hence the name.
From now on, we take $\nu_i < \infty$. The exponential distribution of the holding times leads into the most important of 3 ways to characterize a CTMC.

Three Equivalent Characterizations of CTMC

1) An Embedded Markov Chain in Discrete Time

A Continuous-Time Markov Chain is a stochastic process that moves from state to distinct state according to a Discrete Time Markov Chain, but spends an exponential amount of time in each state. Informally, suppose we are (by Markov WLOG) at state i at time 0, then the embedded chain encodes the probability that the first *other* state the chain will jump to is state j

$$P_{ij} = \mathbb{P}(1^{st} \text{ transition after leaving state } i \text{ is to state } j \mid X(0) = i), i \neq j$$

This is the probability of jumping from state $i \rightarrow j, i \neq j$.

Definition (Embedded Chain)

Let $\{X(t), t \geq 0\}$ be a CTMC with transition rates $\nu_i, i \in \mathcal{S}$. Then, we define the *embedded discrete time Markov chain* $(Y_n)_{n \geq 1}$ as the MC on $\mathcal{S}$ with transition matrix P_{ij} where if $\nu_i > 0$

$$P_{ij} = \mathbb{P}(X(\tau_i) = j \mid X(0) = i), i \neq j$$

and if the state i is absorbing:

$$P_{ii} = 1$$

Note. We should comment on why Y_n is a Markov chain. The reason is simple: it tracks the state of a CTMC, which has the Markov property that conditional on the present, past

is independent of future, which is a property that includes not only the state but also the time, so since the embedded chain contains less information it also has Markov property. P , being the transition matrix of a Markov chain, is a stochastic matrix. If $\nu_i > 0$:

$$\sum_{i \neq j} P_{ij} = 1$$

since $\mathbb{P}(\tau_i = \infty) = 0$ so the chain must jump somewhere! If $\tau_i = 0$, then we stay forever in i . Either way stochastic.

Theorem (CTMC 1st Characterization)

Let state space $\mathcal{S}$. Then, a CTMC $\{X(t), t \geq 0\}$ on $\mathcal{S}$ is completely characterized by the rates $\nu_i, i \in \mathcal{S}$ of the holding times and the embedded chain P_{ij} (along with the initial distribution)

Definition (Rate Matrix)

Let $i, j \in \mathcal{S}$. Then the *transition rate* from $i \rightarrow j$ is stored in the rate matrix $Q = (q_{ij})$:

$$q_{ij} = \begin{cases} \nu_i P_{ij}, & i \neq j \\ -\nu_i, & i = j \end{cases}$$

When we are in state i , since $\tau_i \sim \text{Exp}(\nu_i)$, think of "jump to state $j \neq i$ " as an arrival in a Poisson process, then it makes sense to think of ν_i as the rate at which the process leaves state i and P_{ij} is the probability with which it enters state j given it leaves. It follows then that the transition rate from $i \rightarrow j$ is $\nu_i P_{ij}$

Proposition (Rate Matrix Properties)

The rate matrix of any CTMC satisfies:

1. $\sum_{j \neq i} q_{ij} = \nu_i$
2. $\sum_{j \in \mathcal{S}} q_{ij} = 0$

2) Differential Equations and Infinitesimal Dynamics Approach

The focus of Section 5.4, but I will state the results here for the sake of drilling:

$$P'(t) = P(t)Q, P'(t) = QP(t) \implies P(t) = e^{Qt} = \sum_{n=1}^{\infty} \frac{(tQ)^n}{n!}$$

Thus, the transition function is determined by rate matrix Q .

3) Competing Exponential Clocks

Not covered, but important and perfectly in the scope of the class. Consider being currently in $i \in \mathcal{S}$ and let $j_1, j_2, \dots, j_n$ be the set of states accessible from i . There are k exponential clocks for each of these states, with each one having a timer $T_{i,j_r} \sim \text{Exp}(q_{i,j_r})$. The first timer to go off is the state to which we transition from i . Thus:

$$P_{i,j_m} = \mathbb{P} \left(T_{i,j_m} = \min_{r \in \{1, \dots, n\}} T_{i,j_r} \right) = \frac{q_{i,j_m}}{\sum_{j \neq i} q_{ij}} = \frac{q}{\nu_i}$$

5.3 The Birth and Death Chain

The birth and death chain is on $\mathcal{S} = \{0, 1, 2, \dots\}$ is characterized by birthrate $\lambda_i = q_{i,i+1}$ and death rate $\mu_i = q_{i,i-1}$ and is $q_{ij} = 0$ if $|i - j| > 1$. Thus, the rate matrix is:

$$q_{ij} = \begin{cases} \lambda_i, & \text{if } j = i + 1 \\ \mu_i, & \text{if } j = i - 1 \\ -(\lambda_i + \mu_i) & \text{if } i = j \\ 0, & \text{else} \end{cases}$$

where λ_i is the *birth rate* and μ_i is the *death rate*. The holding time of state i has rate

$$\nu_i = \lambda_i + \mu_i$$

$$P_{i,i+1} = \begin{cases} \frac{\lambda_i}{\lambda_i + \mu_i}, & \text{if } j = i + 1 \\ \frac{\mu_i}{\lambda_i + \mu_i}, & \text{if } j = i - 1 \\ 0, & \text{else} \end{cases}$$

1) M/M/s Queue

Scenario: Queueing, there are s servers who cause people to leave system with rate μ each, and customers arrive at rate λ . All times are exponential, meaning we can add up rates. The birth and death rates:

$$\mu_n = \begin{cases} n\mu & 1 \leq n \leq s \\ s\mu & n > s \end{cases}$$

$$\lambda_n = \lambda, \quad n \geq 0$$

2) Linear Growth with Immigration

Consider a population in which every individual gives birth independently at exponential rate λ and dies at exponential rate μ . New members also enter according to a Poisson process with rate θ . Let $X(t)$ denote the population size at time t . The BD process has rate matrix given by

$$q_{ij} = \begin{cases} i\lambda + \theta, & j = i + 1, \\ i\mu, & j = i - 1, \\ -(i\lambda + i\mu + \theta), & j = i, \\ 0, & \text{otherwise.} \end{cases}$$

where $\lambda_i = q_{i,i+1} = i\lambda + \theta$, $\mu_i = q_{i,i-1} = i\mu$, $\nu_i = i\lambda + i\mu + \theta$. The embedded chain has nonzero entries:

$$P_{i,i+1} = \frac{i\lambda + \theta}{i\lambda + i\mu + \theta}, P_{i,i-1} = \frac{i\mu}{i\lambda + i\mu + \theta}$$

3) Yule Process

Definition (Yule Process)

$\{X(t)\}_{t \geq 0}$ is a CTMC, pure birth process with $\lambda_n = n\lambda$, $n \geq 0$.

Assume $X_0 = 1$ unless otherwise stated.

Note that this is the generalization of branching process to continuous time (n independent Poisson process with rate λ) where arrival times are:

$T_i :=$ time between $(i - 1)$ -st and i th birth = time to transition from $i \rightarrow i + 1$

$\Rightarrow (i - 1)$ -st birth means $X_s = i$, so $\lambda_i = i\lambda$

so $T_i \sim \text{Exp}(i\lambda)$ (i.e. $\tau_i \sim \text{Exp}(\nu_i) = \text{Exp}(i\lambda)$)

Proposition (Yule Process Birth Time Distribution)

Let $S_n = T_1 + \dots + T_n$ with $S_n \sim F_{S_n}$.

Claim: $\mathbb{P}(T_1 + \dots + T_n \leq t) = (1 - e^{-\lambda t})^n$

Note: $\mathbb{P}(T_i \leq t) = 1 - e^{-\lambda t}$

Proof. We compute:

$$\mathbb{P}(T_1 + \dots + T_n \leq t) = \int_0^\infty \mathbb{P}(S + T_n \leq t \mid T_1 + \dots + T_{n-1} = s) dF_{S_{n-1}}(s)$$

Since $T_i \sim \text{Exp}(i\lambda)$:

$$\begin{aligned} &= \int_0^t (1 - e^{-n\lambda(t-s)}) dF_{S_{n-1}}(s) \\ &= \int_0^t dF_{S_{n-1}}(s) - \int_0^t e^{-n\lambda(t-s)} (n-1) e^{-n\lambda s} (1 - e^{-\lambda s})^{n-2} \lambda e^{-\lambda s} ds \end{aligned}$$

Using the substitution $u = 1 - e^{-\lambda s} \Leftrightarrow e^{-\lambda s} = 1 - u$, $\frac{du}{ds} = \lambda e^{-\lambda s}$:

$$\begin{aligned} &= F_{S_{n-1}}(t) - (n-1) e^{-n\lambda t} \int_0^{1-e^{-\lambda t}} (e^{-\lambda s})^n u^{n-2} du \\ &= F_{S_{n-1}}(t) - (n-1) e^{-n\lambda t} \int_0^{1-e^{-\lambda t}} \frac{u^{n-2}}{(1-u)^n} du \\ &= F_{S_{n-1}}(t) - (n-1) e^{-n\lambda t} \int_0^{1-e^{-\lambda t}} \frac{1}{u^1} \left(\frac{u}{1-u} \right)^n du \end{aligned}$$

Let $z = \frac{1}{u} = u^{-1}$, $\frac{dz}{du} = -u^{-2} = -\frac{1}{u^2}$, so $du = dz \cdot u^1$:

$$\begin{aligned} &= F_{S_{n-1}}(t) - (n-1) e^{-n\lambda t} \int_{\frac{1}{1-e^{-\lambda t}}}^{\infty} (z-1)^{-n} dz \\ &= F_{S_{n-1}}(t) - (n-1) e^{-n\lambda t} \left[\frac{1}{1-n} (z-1)^{-n+1} \right]_{\frac{1}{1-e^{-\lambda t}}}^{\infty} \\ &= F_{S_{n-1}}(t) + e^{-n\lambda t} \left(\frac{1}{1-e^{-\lambda t}} - 1 \right)^{-n+1} \\ &= F_{S_{n-1}}(t) + e^{-n\lambda t} \cdot \left(\frac{e^{-\lambda t}}{1-e^{-\lambda t}} \right)^{-n+1} \\ &= F_{S_{n-1}}(t) + e^{-n\lambda t} \cdot \left(\frac{1-e^{-\lambda t}}{e^{-\lambda t}} \right)^{n-1} \end{aligned}$$

Therefore:

$$F_{S_n}(t) = F_{S_{n-1}}(t) \cdot e^{-n\lambda t} \cdot \left(\frac{1-e^{-\lambda t}}{e^{-\lambda t}} \right)^{n-1} = (1 - e^{-\lambda t})^n$$

□

Remark (Key Substitutions Used)

The proof uses the substitution $u = 1 - e^{-\lambda s}$ to transform the exponential integrals, followed by $z = \frac{1}{u}$ to handle the rational function integration.

Theorem (Yule Process Transition Probabilities)

For the Yule process, the transition probabilities are:

$$P_{ij}(t) = \binom{j-1}{i-1} e^{-\lambda t} (1 - e^{-\lambda t})^{j-i}$$

Proof.

$$\mathbb{P}(T_1 + \dots + T_j \leq t) = \mathbb{P}(S_j \leq t) = \mathbb{P}(X_t \geq j+1 \mid X_0 = 1) \quad (5)$$

$$\Rightarrow P_{1j}(t) = \mathbb{P}(X_t \geq j \mid X_0 = 1) - \mathbb{P}(X_t \geq j+1 \mid X_0 = 1) \quad (6)$$

$$= (1 - e^{-\lambda t})^{j-1} - (1 - e^{-\lambda t})^j \quad (7)$$

$$= (1 - e^{-\lambda t})^{j-1} (1 - 1 + e^{-\lambda t}) \quad (8)$$

$$= (1 - e^{-\lambda t})^{j-1} e^{-\lambda t} \sim \text{Geom}(e^{-\lambda t}) \quad (9)$$

For P_{ij} we have sum of i i.i.d. $\text{Geom}(e^{-\lambda t}) \Rightarrow \text{NB}(e^{-\lambda t})$ (kind of)

Yule process is pure birth so $P_{ij}(t) = 0$ if $i > j$, thus we must shift the negative binomial distribution:

Therefore:

$$P_{ij}(t) = \binom{j-1}{i-1} e^{-\lambda t} (1 - e^{-\lambda t})^{j-i} \quad (\text{Yule Transition Probabilities})$$

□

5.4 Kolmogorov Differential Equations

Key Lemmas

Theorem

Suppose $X(0) = i$, then

1. $\mathbb{P}(0 \text{ transitions in } (0, h]) = 1 - \nu_i h + o(h)$
2. $\mathbb{P}(\text{exactly 1 transition in } (0, h]) = \nu_i h + o(h)$
3. $\mathbb{P}(\geq 2 \text{ transitions in } (0, h]) = o(h)$

Proof. Note that these probabilities sum to 1, since there must be some finite number of transitions by regularity. (1) is easy:

$$\mathbb{P}(0 \text{ transitions in } (0, h]) = \mathbb{P}(\tau_i > h) = e^{-\nu_i h} = 1 - \nu_i h + o(h)$$

To finish the proof, we show (3)

Assume $X(0) = i$. The event $E = \{2 \text{ or more transitions in } (0, h]\} = \bigcup_{j \in S, j \neq i} E_j$ where $E_j = \{\tau_i < h, \text{ 1st transition after leaving state } i \text{ is to } j, \tau_i + \tau_j < h\}$. If $\nu_i \neq \nu_j$ for any $j \in S$ we have:

$$\begin{aligned}
\mathbb{P}(E_j) &= \mathbb{E}(\mathbb{1}_{E_j}) \\
&= \mathbb{E}(\mathbb{E}(\mathbb{1}_{E_j} \mid \tau_i)) \\
&= \int_0^h \mathbb{P}(E_j \mid \tau_i = t) \nu_i e^{-\nu_i t} dt \\
&= \int_0^h P_{ij} \mathbb{P}(\tau_j < h - t) \nu_i e^{-\nu_i t} dt \\
&= q_{ij} \int_0^h (1 - e^{-\nu_j(h-t)}) e^{-\nu_i t} dt \\
&= q_{ij} \int_0^h (e^{-\nu_i t} - e^{-\nu_j h} e^{(\nu_j - \nu_i)t}) dt \\
&= q_{ij} \left[\frac{-1}{\nu_i} e^{-\nu_i t} - \frac{e^{-\nu_j h}}{\nu_j - \nu_i} e^{(\nu_j - \nu_i)t} \right]_0^h \\
&= -q_{ij} \left(\frac{1}{\nu_i} e^{-\nu_i h} + \frac{e^{-\nu_j h}}{\nu_j - \nu_i} e^{(\nu_j - \nu_i)h} - \frac{1}{\nu_i} - \frac{e^{-\nu_j h}}{\nu_j - \nu_i} \right) \\
&= -q_{ij} \left(\frac{1}{\nu_i} (e^{-\nu_i h} - 1) + \frac{1}{\nu_j - \nu_i} (e^{-\nu_i h} - e^{-\nu_j h}) \right) \\
&= -q_{ij} \left(\frac{1}{\nu_i} (1 - \nu_i h + o(h) - 1) + \frac{1}{\nu_j - \nu_i} [1 - \nu_i h + o(h) - 1 + \nu_j h + o(h)] \right) \\
&= -q_{ij} (-h + o(h) + \frac{1}{\nu_j - \nu_i} [(\nu_j - \nu_i)h + o(h)]) \\
&= -q_{ij} (-h + o(h) + h + o(h)) \\
&= -q_{ij} o(h)
\end{aligned}$$

Now suppose $\exists j \in S, \nu_j = \nu_i$, then the integral is slightly simpler:

$$\begin{aligned}
\mathbb{P}(E_j) &= q_{ij} \int_0^h (e^{-\nu_i t} - e^{-\nu_j h}) dt \\
&= q_{ij} \left[\frac{-1}{\nu_i} e^{-\nu_i t} - t e^{-\nu_j h} \right]_0^h \\
&= -q_{ij} \left[\frac{1}{\nu_i} e^{-\nu_i h} + h e^{-\nu_j h} - \frac{1}{\nu_i} \right] \\
&= -q_{ij} \left[\frac{1}{\nu_i} (1 - \nu_i h + o(h)) + h(1 - \nu_j h + o(h)) - \frac{1}{\nu_i} \right] \\
&= -q_{ij} [-h + h - \nu_j h^2 + o(h)] \\
&= -q_{ij} o(h)
\end{aligned}$$

$$\implies \mathbb{P}(E \mid X_0 = i) = \sum_{j \in S, j \neq i} \mathbb{P}(E_j \mid X_0 = i) = - \sum_{j \neq i} q_{ij} o(h) = -\nu_i o(h) = o(h)$$

since $\nu_i < \infty$. □

Lemma

For a CTMC:

1. $\lim_{h \rightarrow 0} \frac{1 - P_{ii}(h)}{h} = \nu_i$
2. $\lim_{h \rightarrow 0} \frac{P_{ij}(h)}{h} = q_{ij}, i \neq j$

Proof. (1) $1 - P_{ii}(h) = \sum_{j \neq i} P_{ij}(h) = \mathbb{P}(\geq 1 \text{ transitions in } (0, h]) = \nu_i h + o(h) \implies \lim_{h \rightarrow 0} \frac{\nu_i h + o(h)}{h} = \nu_i$

(2) In concept, this is easy, but a little trickier to prove. Key is conditioning on the number of jumps in the interval and using the previous result:

$$\begin{aligned}
P_{ij}(h) &= \mathbb{P}(X(h) = j \mid X(0) = i) \\
&= \mathbb{P}(X(h) = j \mid X(0) = i, 1 \text{ transition in } (0, h]) \mathbb{P}(1 \text{ transition in } (0, h] \mid X(0) = i) \\
&+ \mathbb{P}(X(h) = j \mid X(0) = i, \geq 2 \text{ transitions in } (0, h]) \mathbb{P}(\geq 2 \text{ transitions in } (0, h] \mid X(0) = i) \\
&= P_{ij}(\nu_i h + o(h)) + o(h) \\
&= \frac{q_{ij}}{\nu_i} \nu_i h + o(h) \\
&= q_{ij} h + o(h)
\end{aligned}$$

□

Chapman-Kolmogorov

Proposition (Chapman-Kolmogorov Equation)

For any $s, t \geq 0$ we have:

$$P_{ij}(t+s) = \sum_{k \in \mathcal{S}} P_{ik}(t) P_{kj}(s)$$

and in matrix form noting that it may be infinite:

$$P(t+s) = P(t)P(s)$$

Proof. This is easy:

$$\begin{aligned} P_{ij}(s+t) &= \mathbb{P}(X(t+s) = j \mid X(0) = i) \\ &= \sum_{k \in \mathcal{S}} \mathbb{P}(X(t+s) = j \mid X(s) = k, X(0) = i) \mathbb{P}(X(s) = k \mid X(0) = i) \\ &= \sum_{k \in \mathcal{S}} \mathbb{P}(X(t+s) = j \mid X(s) = k) P_{ik}(s) \\ &= \sum_{k \in \mathcal{S}} P_{ik}(s) P_{kj}(t) \end{aligned}$$

I did it the other way, but whatever it still implies the result. □

Kolmogorov Backward and Forward Equations

Theorem (Kolmogorov Backward Equations)

$$P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - \nu_i P_{ij}(t)$$

In matrix form:

$$P'(t) = QP(t)$$

$$\implies P_{ij}(t+h) = P_{ij}(h+t) = \sum_{k \in \mathcal{S}} P_{ik}(h) P_{kj}(t)$$

$$\implies P_{ij}(t+h) - P_{ij}(t) = \sum_{k \neq i} P_{ik}(h) P_{kj}(t) + P_{ii}(h) P_{ij} - P_{ij}(t)$$

$$\begin{aligned}
&= \sum_{k \neq i} P_{ik}(h)P_{kj}(t) - P_{ij}(t)(1 - P_{ii}(h)) \\
\Rightarrow \lim_{h \rightarrow 0} \frac{P_{ij}(t+h) - P_{ij}(t)}{h} &= \sum_{k \neq i} \lim_{h \rightarrow 0} \frac{P_{ik}(h)}{h} P_{kj}(t) - \lim_{h \rightarrow 0} \frac{P_{ij}(t)(1 - P_{ii}(h))}{h} \\
&= \sum_{k \neq i} q_{ik}P_{kj}(t) - \nu_i P_{ij}(t)
\end{aligned}$$

Theorem (Kolmogorov Forward Equations)

Let $P(t)$ be the transition matrix of a CTMC with generator Q . Then the Kolmogorov Forward Equations are (in summation form):

$$P'_{ij}(t) = \sum_{k \neq j} P_{ik}(t)q_{kj}(t) - P_{ij}(t)\nu_j$$

and in matrix form:

$$P'(t) = P(t)Q$$

$$\begin{aligned}
\Rightarrow P_{ij}(t+h) - P_{ij}(t) &= \sum_{k \in \mathcal{S}} P_{ik}(t)P_{kj}(h) - P_{ij}(t) \\
&= \sum_{k \neq j} P_{ik}(t)P_{kj}(h) + P_{ij}(t)P_{jj}(h) - P_{ij}(t) \\
&= \sum_{k \neq j} P_{ik}(t)P_{kj}(h) - P_{ij}(t)[1 - P_{jj}(h)] \\
\Rightarrow \lim_{h \rightarrow 0} \frac{P_{ij}(t+h) - P_{ij}(t)}{h} &= \lim_{h \rightarrow 0} \frac{\sum_{k \neq j} P_{ik}(t)P_{kj}(h) - P_{ij}(t)[1 - P_{jj}(h)]}{h} \\
&= \sum_{k \neq j} P_{ik}(t) \lim_{h \rightarrow 0} \frac{P_{kj}(h)}{h} - P_{ij}(t) \lim_{h \rightarrow 0} \frac{1 - P_{jj}(h)}{h} \\
&= \sum_{k \neq j} P_{ik}(t)q_{kj} - P_{ij}(t)\nu_j \quad \square
\end{aligned}$$

Remark (Kolmogorov Initial Condition)

$$P(0) = I$$

Remark (Remembering KFE and KBE)

Backward equation has Q behind $P(t)$, forward equation has Q in front of $P(t)$

Theorem (Kolmogorov Differential Equations and Matrix Exponential Solution)

Let $\{X(t), t \geq 0\}$ be a CTMC with rate matrix $Q = (q_{ij})$ and transition function $P(t) = (P_{ij}(t))$. Then:

Kolmogorov Forward Equations:

$$\frac{d}{dt}P(t) = P(t)Q$$

Kolmogorov Backward Equations:

$$\frac{d}{dt}P(t) = QP(t)$$

with initial condition $P(0) = I$ (identity matrix).

Matrix Exponential Solution: The unique solution to both differential equations is:

$$P(t) = e^{Qt} = \sum_{n=0}^{\infty} \frac{(tQ)^n}{n!}$$

where $Q^0 = I$ by convention.

Consequence: The transition function is completely determined by the rate matrix Q .

This is why Q is often called the generator matrix

5.5 Stationary Distributions and Long Run Behavior

We did not cover this in lecture, but it was on the HW, so I will briefly state the basics, focusing on proving the ergodic theorem using renewal theory. Based on Section 4.8 but stated in CTMC context.

Definition (Irreducible CTMC)

A CTMC is irreducible $\iff$ the embedded chain $(Y_n)_{n \geq 1}$ irreducible

We assume CTMC we discuss irreducible.

Definition (Return Time)

$T_{ii} :=$ time to return to state i , given that $X(0) = i$

Definition (Mean Return Time)

$$m_i = \mathbb{E}(T_{ii})$$

Definition (Positive Recurrence)

A CTMC is positive recurrent $\iff m_i < \infty, \forall i \in \mathcal{S}$

Theorem

If a CTMC is irreducible and positive recurrent, then

$$P_j = \lim_{t \rightarrow \infty} \mathbb{P}(X_t = j \mid X_0 = i) = \lim_{t \rightarrow \infty} P_{ij}(t)$$

exists and is finite AND independent of initial state i

Pf Sketch. Define 'on' as when the CTMC is in state i and "off" otherwise. This is an alternating renewal process. The cycles are i.i.d. by the structure of the CTMC (exponential holding times and defined P_{ij}). The alternating renewal requires that $\mathbb{E}(Z_i + Y_i) < \infty$ where $Z_i + Y_i$ is the time between successive transitions into "on" state (i), thus $m_i < \infty$ is exactly this condition. Then, ART $\implies P_j = \frac{\mu_j}{m_j} = \frac{1}{\nu_j m_j}$ $\square$

Definition

An irreducible, positive recurrent CTMC is said to be ergodic

Proposition (Limiting Probability Sufficient Condition)

If a Continuous-Time Markov Chain is irreducible and if T_{ii} has a non-lattice distribution with $\mathbb{E}(T_{ii}) < \infty$ (i.e. ergodic), then the limiting probability exists and is:

$$P_j = \frac{\pi_j}{\nu_j \sum_{i \in \mathcal{S}} \frac{\pi_i}{\nu_i}}$$

where π is the unique stationary distribution of the embedded Markov Chain

We can define a row vector of long-run proportions as:

$$\alpha_j = \mathbb{P}(X(t) = j)$$

This only really makes sense if limit probability exists

Definition (Stationary Distribution)

A row vector π satisfying

$$\alpha Q = \mathbf{0}, \quad \sum_i \alpha_i = 1, \quad \alpha_i \geq 0,$$

is a *stationary distribution*. If it exists and the chain is irreducible, then $\alpha P(t) = \alpha$ for all $t \geq 0$.

It can be shown that the 2 notions of stationarity $\vec{\alpha}Q = 0, \alpha P(t) = \alpha$ are equivalent

Theorem (CTMC Ergodic Theorem)

An irreducible CTMC is ergodic $\iff$ the above equations have a unique solution

This can be proved using renewal theory.

Corollary (Reversible Chain and Detailed Balance)

If a CTMC is reversible with distribution π , then for all i, j

$$\pi_i q_{ij} = \pi_j q_{ji}.$$

Theorem (Stationary Distribution of Birth/Death Chain)

For an irreducible birth and death chain the stationary distribution is:

$$P_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n (1 + S)}, n \geq 1$$

$$P_0 = \frac{1}{1 + S}$$

$$S = \sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n}$$

and also $S < \infty \iff$ chain ergodic (limiting probability exists)

Chapter 6: Martingales

6.0 Conditional Expectation, Supercharged

This section requires a taste of graduate probability. The professor repeatedly mentioned "the smaller sigma field wins". To explain this begin with the following definition:

Definition (Conditional Expectation on Sigma Field)

Let $(\Omega, \mathcal{F}_o, \mathbb{P})$ be a probability space and let $\mathcal{F} \subset \mathcal{F}_o$ be sigma-algebra on Ω . Then, the *conditional expectation of random variable X given $\mathcal{F}$* , denoted $\mathbb{E}(X | \mathcal{F})$, is defined as the unique random variable Y satisfying:

1. $Y \in \mathcal{F}$
2. For all $A \in \mathcal{F}$, $\int_A X dP = \int_A Y dP$

Note: In practice, this reduces to the familiar formulas:

- Discrete: $E[X|Y = y] = \sum_x x \cdot P(X = x|Y = y)$
- Continuous: $E[X|Y = y] = \int x \cdot f_{X|Y}(x|y) dx$

Definition (Conditional Expectation on RV)

For random variables X, Y , the conditional expectation of X given Y , denoted $\mathbb{E}(X | Y)$ is $\mathbb{E}(X | \sigma(Y))$

6.1 Martingale Basics

The martingale is a generalization of a "fair game." By fair game, one usually means that in expectation, neither the house or player profits. In finance, this is essentially the "principle of no arbitrage." Let us walk through several definitions. The first comes from the course text, Ross. The second, a more rigorous version involving measure theory, that I find illustrative

1) Ross's Elementary Definition

Definition (Martingale)

A stochastic process $\{Z_n, n \geq 1\}$ is a *martingale* if:

$$\mathbb{E}(|Z_n|) < \infty$$

and

$$\mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n) = Z_n$$

for all $n \geq 1$

2) A Measure Theory Framework

Definition (Filtration)

A *filtration* $\{\mathcal{F}_n\}_{n \geq 0}$ is an increasing sequence of σ -fields, meaning:

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$$

Note. For example for a sequence of random variables $(\mathcal{F}_n)_{n \geq 1}$, where $\mathcal{F}_n = \sigma(Z_1, \dots, Z_n)$. Here "increasing" refers to containment, not numerical order. Intuitively, $\mathcal{F}_n$ represents the information available at time n .

Definition (Martingale II)

Let $\{\mathcal{F}_n\}$ be a filtration. A sequence $\{X_n\}$ is said to be a *martingale* with respect to $\{\mathcal{F}_n\}$ if:

1. $E|X_n| < \infty$ for all n (integrability)
2. $X_n \in \mathcal{F}_n$ for all n (X_n adapted to $\mathcal{F}_n$)
3. $E[X_{n+1} | \mathcal{F}_n] = X_n$ for all n (martingale property)

If condition (3) is replaced by $E[X_{n+1} | \mathcal{F}_n] \leq X_n$, then $\{X_n\}$ is called a *supermartingale*.

If condition (3) is replaced by $E[X_{n+1} | \mathcal{F}_n] \geq X_n$, then $\{X_n\}$ is called a *submartingale*.

Note. Condition (2) means that not only can a martingale be define w.r.t. $\sigma(Z_1, \dots, Z_n)$ (the so-called "natural filtration") but also any larger σ -field (i.e. any field containing more information)

Proposition (Constant Expectation of Martingales)

If $\{Z_n, n \geq 1\}$ is a martingale, then

$$\mathbb{E}(Z_{n+1}) = \mathbb{E}(Z_n)$$

Proof. This is easy, just take expectations of both sides of the martingale property and use the tower property. $\square$

Remark (Nested Tower Property)

We have for a random vectors $\vec{U}, \vec{Y}$:

$$\mathbb{E}(X | \vec{U}) = \mathbb{E}\left(\mathbb{E}(X | \vec{Y}, \vec{U}) | \vec{U}\right)$$

Note. the smaller sigma field wins (i.e. always condition additionally on the thing which gives you the most information)

3) Some Easy Examples

1. $(X_k)_{k \geq 1}$ independent with 0 mean and let $Z_n = \sum_{i=1}^n X_i$; then $(Z_n)_{n \geq 1}$ is a martingale since:

$$\begin{aligned}\mathbb{E}(Z_{n+1} | Z_1, \dots, Z_n) &= \mathbb{E}(Z_n + X_{n+1} | Z_1, \dots, Z_n) \\ &= \mathbb{E}(Z_n | Z_1, \dots, Z_n) + \mathbb{E}(X_{n+1} | Z_1, \dots, Z_n) \\ &= Z_n + \mathbb{E}(X_{n+1}) = Z_n\end{aligned}$$

2. $(X_k)_{k \geq 1}$ independent with $\mathbb{E}(X_i) = 1$, then $(Z_n)_{n \geq 1}$ is a martingale when $Z_n = \prod_{i=1}^n X_i$.

To see this:

$$\begin{aligned}\mathbb{E}(Z_{n+1} | Z_1, \dots, Z_n) &= \mathbb{E}(X_{n+1} Z_n | Z_1, \dots, Z_n) \\ &= \mathbb{E}(X_{n+1} | Z_1, \dots, Z_n) \mathbb{E}(Z_n | Z_1, \dots, Z_n) \\ &= \mathbb{E}(X_{n+1}) Z_n \\ &= Z_n\end{aligned}$$

Note: $Z_n = g(X_1, \dots, X_n) \perp X_{n+1}$, $(Z_1, \dots, Z_n) = h(X_1, \dots, X_n) \perp X_{n+1}$, $\mathbb{E}(X_{n+1}) = 1$.

3. Consider a branching process, let X_n = size of n th generation and $m = \mathbb{E}(Z)$, $Z \leftarrow$ offspring RV. Then $Z_n = \frac{X_n}{m^n}$ is a martingale if $n \geq 1$.

$$\begin{aligned}\mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n) &= \mathbb{E} \left[\frac{X_{n+1}}{m^{n+1}} \mid Z_1, \dots, Z_n \right] \\ &= \frac{1}{m^{n+1}} \mathbb{E}(X_{n+1} \mid X_n) \\ &= \frac{mX_n}{m^{n+1}} = \frac{X_n}{m^n} = Z_n\end{aligned}$$

Remark

If we can show $\mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n, \vec{Y}) = Z_n, \implies \mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n) = Z_n$.

Proof. Suppose $\mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n, \vec{Y}) = Z_n$, then

$$\begin{aligned}\mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n) &= \mathbb{E} \left[\mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n, \vec{Y}) \mid Z_1, \dots, Z_n \right] \\ &= \mathbb{E}(Z_n \mid Z_1, \dots, Z_n) = Z_n\end{aligned}$$

That is, if martingale property satisfied on bigger sigma field, then we are good

□

4. Let $X, Y_1, Y_2, \dots$ be arbitrary RVs such that $\mathbb{E}(|X|) < \infty$ and let

$$Z_n = \mathbb{E}(X \mid Y_1, \dots, Y_n)$$

We find:

$$\begin{aligned}\mathbb{E}(Z_{n+1} \mid Y_1, \dots, Y_n) &= \mathbb{E}(\mathbb{E}(X \mid Y_1, \dots, Y_{n+1}) \mid Y_1, \dots, Y_n) \\ &= \mathbb{E}(X \mid Y_1, \dots, Y_n) \\ &= Z_n\end{aligned}$$

$$\begin{aligned}\mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n) &= \mathbb{E}(\mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n, Y_1, \dots, Y_n) \mid Z_1, \dots, Z_n) \\ &= \mathbb{E}(Z_n \mid Z_1, \dots, Z_n) \\ &= Z_n\end{aligned}$$

5. Generalization of the fact that the partial sum of independent mean 0 RVs is a martingale.

Let $X_1, X_2, \dots$ be arbitrary RVs, then $X_i - \mathbb{E}(X_i \mid X_1, \dots, X_{i-1})$ have mean 0. [need not be independent]

$$Z_n := \sum_{i=1}^n [X_i - \mathbb{E}(X_i \mid X_1, \dots, X_{i-1})] = g(X_1, \dots, X_n)$$

Then if $\mathbb{E}(|Z_n|) < \infty$, $(Z_n)_{n \geq 1}$ is a martingale with mean 0.

To see this note that $Z_{n+1} = Z_n + X_{n+1} - \mathbb{E}(X_{n+1} \mid X_1, \dots, X_n)$

$$\begin{aligned} \mathbb{E}(Z_{n+1} \mid Z_1, \dots, Z_n) &= \mathbb{E}(Z_n + X_{n+1} - \mathbb{E}(X_{n+1} \mid X_1, \dots, X_n) \mid Z_1, \dots, Z_n) \quad (10) \\ &= Z_n + \mathbb{E}(X_{n+1} \mid Z_1, \dots, Z_n) - \mathbb{E}(\mathbb{E}(X_{n+1} \mid X_1, \dots, X_n, Z_1, \dots, Z_n) \mid Z_1, \dots, Z_n) \\ &= Z_n \end{aligned}$$

6.2 Martingale Random Times, Stopping Times

Definition (Random Time)

Let $N : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ be a random variable. N is a *random time* for the sequence of random variables $(Z_k)_{k \geq 1}$ if the event $\{N = n\}$ can be determined by $Z_1, \dots, Z_n$

Remark (Measure Theory Definition)

Let $N : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ be a random variable and $(\mathcal{F}_n)_{n \geq 1}$ be the natural filtration where $\mathcal{F}_n = \sigma(Z_1, \dots, Z_n)$. Then N is a *stopping time* (or *random time*) with respect to the filtration $(\mathcal{F}_n)$ if:

$$\{N = n\} \in \mathcal{F}_n \text{ for all } n \in \mathbb{N}$$

Note (Intuitive Meaning). The condition $\{N = n\} \in \mathcal{F}_n = \sigma(Z_1, \dots, Z_n)$ means that we can determine whether $N = n$ with only the information available from $Z_1, \dots, Z_n$ (i.e. available info up to time n).

Definition (Stopping Time)

An almost surely finite random time

Definition (Stopped Process $\bar{Z}_n$)

$$\bar{Z}_n = \begin{cases} Z_n, & \text{if } n \leq N \\ Z_N, & \text{if } n > N \end{cases}$$

This is the process stuck at whatever value it takes at stopping time N

Proposition

For N a random time for $(Z_n)_{n \geq 1}$, $\{\bar{Z}_n\}_{n \geq 1}$ (stopped process) is a martingale.

Proof. Let $I_n = \begin{cases} 1 & \text{if } N \geq n \\ 0 & \text{if } N < n \end{cases} = g(Z_1, \dots, Z_{n-1})$ where g indicates whether or not the process stopped by $n - 1$.

Claim: $\bar{Z}_n = \bar{Z}_{n-1} + I_n(Z_n - Z_{n-1})$

For $n \leq N$, then $\bar{Z}_n = Z_n = Z_{n-1} + Z_n - Z_{n-1}$

For $n > N$, then $\bar{Z}_n = Z_N = Z_{n-1} + 0$

where $n \geq N + 1$ and $N - 1 \geq N$.

Thus,

$$\begin{aligned} \mathbb{E}(\bar{Z}_n \mid Z_1, \dots, Z_{n-1}) &= \mathbb{E}[\bar{Z}_{n-1} + I_n(Z_n - Z_{n-1}) \mid Z_1, \dots, Z_{n-1}] \\ &= \mathbb{E}[\bar{Z}_{n-1} \mid Z_1, \dots, Z_{n-1}] + I_n \mathbb{E}[Z_n \mid Z_1, \dots, Z_{n-1}] - I_n Z_{n-1} \end{aligned}$$

If we know $Z_1, \dots, Z_{n-1}$ then we know whether or not $(Z_n)_{n \geq 1}$ has stopped up to $n - 1$, so we know if $N \geq n$, so we know $\bar{Z}_1, \dots, \bar{Z}_{n-1}$.

“Smaller σ -field wins” - condition additionally on more informative RVs:

$$\begin{aligned} \mathbb{E}[\bar{Z}_n \mid \bar{Z}_1, \dots, \bar{Z}_{n-1}] &= \mathbb{E}[\mathbb{E}(\bar{Z}_n \mid \bar{Z}_1, \dots, \bar{Z}_{n-1}, Z_1, \dots, Z_{n-1}) \mid \bar{Z}_1, \dots, \bar{Z}_{n-1}] \\ &= \mathbb{E}(\bar{Z}_{n-1} \mid \bar{Z}_1, \dots, \bar{Z}_{n-1}) \\ &= \bar{Z}_{n-1} \end{aligned}$$

□

Theorem (Martingale Stopping Theorem)

If $(Z_n)_{n \geq 1}$ is a martingale on $\mathcal{F}_n$ and N is a stopping time for the same sequence $\mathcal{F}_n$ and ANY of the following hold:

1. The stopped martingale $(\bar{Z}_n)_{n \geq 1}$ is uniformly bounded: $\exists M \in \mathbb{R}, |\bar{Z}_n| \leq M, \forall n \in \mathbb{N}$
2. N is bounded: $\exists B \in \mathbb{N}, N \leq B$ w.p. 1
3. $\mathbb{E}(N) < \infty$ and $\exists M' < \infty$ such that:

$$\mathbb{E}(|Z_{n+1} - Z_n| \mid \mathcal{F}_n) \leq M'$$

then

$$\mathbb{E}(Z_N) = \mathbb{E}(Z_1)$$

Theorem (Martingale Convergence Theorem)

If $(Z_n)_{n \geq 1}$ is a martingale such that for some $M < \infty$, $\mathbb{E}(|Z_n|) \leq M, \forall n \in \mathbb{N}$ then:

$$\lim_{n \rightarrow \infty} Z_n$$

exists and is finite with probability 1.

Chapter 8: Brownian Motion

Motivations: The Limit of a Random Walk

The text starts with a very long, but very informal description of how to construct a Brownian motion. The summary is that Brownian Motion is a type of limit of a random walk, where we take the sum of a sequence of i.i.d. mean 0, variance 1 RVs, but scale by the step size Δx . We divide the continuous time interval $[0, t]$ into $N = \lfloor \frac{t}{\Delta t} \rfloor$ segments. Steps are only allowed at integer multiples of Δt and are of size Δx .

$$X_N(t) = \Delta x(X_1 + \dots + X_N)$$

$$\text{where } X_i = \begin{cases} 1, & \text{w.p. } \frac{1}{2} \\ -1, & \text{w.p. } \frac{1}{2} \end{cases}$$

we take $\Delta x = c\sqrt{\Delta t}$ and let $\Delta t \rightarrow 0$, then $N \approx \frac{t}{\Delta t} \rightarrow \infty$ and by CLT we have

$$X_t = \lim_{\Delta t \rightarrow 0} X_N(t) = \lim_{N \rightarrow \infty} c\sqrt{\frac{t}{N}}(X_1 + \dots + X_N) \sim c\sqrt{t}\mathcal{N}(0, 1) \sim \mathcal{N}(0, c^2t)$$

The fact that BM can be thought of/constructed this way shows that the process has stationary increments and independent increments simply because it is a limit of a sum of i.i.d. guys.

8.1) Fundamentals

Definition 1

Definition (Brownian Motion)

A stochastic process $(X_t)_{t \geq 0}$ is said to be a Brownian motion process if:

1. $X(0) = 0$
2. $(X_t)_{t \geq 0}$ has *stationary increments*
3. $(X_t)_{t \geq 0}$ has *independent increments*
4. $X(t) \sim \mathcal{N}(0, c^2t), \forall t > 0$

Note. For $c = 1$, we have the *standard Brownian motion* and since $X(t)/c, c > 0$ is always a standard Brownian motion we take WLOG $c = 1$ from now on.

$X(t) - X(s) :=$ increment and concerns the position (think stock price) in the half-open interval $(0, t]$

- $X(t)$ is known as the sample path and *must* be continuous
- With probability 1, $X(t)$ is nowhere differentiable
- The density function of $X(t)$, denoted $f_t(x) = \frac{1}{\sqrt{2\pi t}}e^{-\frac{x^2}{2t}}$

The following theorem is a consequence of the independent increments assumption.

Theorem (Brownian Motion is a Markov Process)

Brownian motion is a *Markov process*. This means that:

$$\mathbb{P}(X(t+s) \leq a \mid X(s) = x, X(u), 0 \leq u < s) = \mathbb{P}(X(t+s) \leq a \mid X(s) = x)$$

(any process satisfying that is a Markov process, but we are interested in Brownian motion)

In words, this means that the conditional distribution of a future state given the entire past history and the present depends only on the present

Proof. The calculation presented without comment:

$$\begin{aligned}\mathbb{P}(X(t+s) \leq a \mid X(s) = x, X(u), 0 \leq u < s) &= \mathbb{P}(X(t+s) - X(s) \leq a - x \mid X(s) = x, X(u), 0 \leq u < s) \\ &= \mathbb{P}(X(t+s) - X(s) \leq a - x \mid X(s) = x) \\ &= \mathbb{P}(X(t+s) \leq a \mid X(s) = x)\end{aligned}$$

Not an obvious proof. We explain step-by-step. The first step is algebra. Subtract x from both sides of the inequality to obtain $X(t+s) - x \leq a - x$, but since we condition on $X(s) = x$, this is $X(t+s) - X(s) \leq a - x$. By independent increments we can get rid of everything up to and including $X(s)$, but we leave $X(s)$ by choice $\mathbb{P}(X(t+s) - X(s) \leq a - x)$, then add x to both sides again to obtain the result. $\square$

Proposition ($X(t)$ Density Function)

The density function of $X(t) \sim \mathcal{N}(0, t)$ is given by

$$f_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$$

Note (Symmetry of Brownian Motion). Because the distribution of X_t is normal, we have

$$-X_t \stackrel{d}{=} X_t$$

(if you plug in $-x$ to pdf you get same thing because even function)

Remark (Independent and Stationary Increments Reminder)

For $0 = t_0 < t_1 < \dots < t_n$, $\{X(t_i) - X(t_{i-1})\}_{i \geq 1}$ form an independent sequence (i.e. increments on disjoint intervals independent)

For $0 \leq t_1 < t_2, s \geq 0$, The distribution of $X(t_2 + s) - X(t_1 + s)$ does not depend on s , only the length of the interval or equivalently:

$$\forall s_1, s_2 \geq 0, X(t_2 + s_1) - X(t_1 + s_1) \stackrel{d}{=} X(t_2 + s_2) - X(t_1 + s_2)$$

Stationary increments corresponds to shift invariance

Proposition

For Brownian motion $\{X(t), t \geq 0\}$ and $s < t$, we have: $X(t) - X(s) \sim X(t - s)$

Proof. This is trivial. By stationary increments

$$X((t-s)+s) - X(0+s) \stackrel{d}{=} X(t-s) - X(0) = X(t-s) \sim \mathcal{N}(0, t-s)$$

Alternatively, note $X_t = X_s + (X_t - X_s)$. By Independent Increments, $X_s \perp (X_t - X_s)$ and by definition $X_t \sim \mathcal{N}(0, t)$, $X_s \sim \mathcal{N}(0, s)$. By Levy-Cramer Theorem, if $X + Y$ normal then X and Y are both normal. Thus $Y = X_t - X_s = \mathcal{N}(\mu, \sigma^2)$. A normal distribution is completely characterized by mean and variance. So result follows if we compute mean and variance.

$$\mathbb{E}(X_t) = \mathbb{E}(X_s) + \mathbb{E}(X_t - X_s) \implies \mathbb{E}(X_t - X_s) = 0$$

$$\text{Var}(X_t) = \text{Var}(X_s) + \text{Var}(X_t - X_s) \implies \text{Var}(X_t - X_s) = t - s$$

□

Theorem (Joint Density of Brownian Motion)

Let $t_1, \dots, t_n \in [0, \infty)$ be arbitrary times. The joint density function of $X(t_1), \dots, X(t_n)$ is given by

$$f(x_1, x_2, \dots, x_n) = f_{t_1}(x_1) f_{t_2-t_1}(x_2 - x_1) \dots f_{t_n-t_{n-1}}(x_n - x_{n-1})$$

Proof. It is a rather deep and shocking fact that joint density are just like intersections of probabilities.

Let's just compute the joint probability:

$$\mathbb{P}(X(t_1) = x_1, \dots, X(t_n) = x_n) = \mathbb{P}(X(t_1) = x_1) \prod_{k=2}^n \mathbb{P}(X(t_k) = x_k \mid X(t_i) = x_i, 1 \leq i < k)$$

Similar to the Markov proof, we are conditioning on $X(t_{k-1}) = x_{k-1}$, thus the probability inside the pi product can be written:

$$\mathbb{P}(X(t_k) = x_k \mid X(t_i) = x_i, 1 \leq i < k) = \mathbb{P}(X(t_k) - X(t_{k-1}) = x_k - x_{k-1} \mid X(t_i) = x_i, 1 \leq i < k)$$

By independent increments $X(t_k) - X(t_{k-1}) \perp X(t_1), \dots, X(t_{k-1})$ and by previous proposition (stationary increments) $X(t_k) - X(t_{k-1}) \sim X(t_k - t_{k-1})$. Thus, the intersection is:

$$\mathbb{P}(X(t_1) = x_1) \prod_{k=2}^n \mathbb{P}(X(t_k - t_{k-1}) = x_k - x_{k-1})$$

It is worth writing out the subscripts of the density function:

$$f = f_{X(t_1), \dots, X(t_n)}$$

and when we input arguments we take $X(t_i) = x_i$!

$$\implies f(x_1, \dots, x_n) = f_{t_1}(x_1) f_{t_2-t_1}(x_2) \cdots f_{t_n-t_{n-1}}(x_n)$$

□

Corollary (Explicit Form of Joint Density)

For a Brownian Motion $\{X(t), t \geq 0\}$ and times $0 < t_1 < t_2 < \cdots < t_n$, we have

$$f_{X(t_1), X(t_2), \dots, X(t_n)}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} \exp\left(-\frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})}\right) \quad (11)$$

where we define $t_0 = 0$ and $x_0 = 0$.

Definition (Gaussian process)

A stochastic process $\{X(t), t \geq 0\}$ is called a *Gaussian process* if $X(t_1), \dots, X(t_n)$ has a multivariate normal distribution for any arbitrary collection of times $t_1, \dots, t_n \in [0, \infty)$

This definition says: pick some finite collection of times get a MVN.

Definition 2: MVN/Gaussian Process

Definition (Brownian Motion II)

A continuous-time stochastic process is Brownian motion iff:

1. $X(0) = 0$
2. $X(t)$ continuous for all $t \geq 0$
3. $\{X(t), t \geq 0\}$ is a Gaussian process with

$$\mathbb{E}(X(t)) = 0$$

and

$$\text{Cov}(s, t) = \min(s, t)$$

Theorem

Brownian motion is a Gaussian process.

Proof. Let $\{X_t, t \geq 0\}$ be a Brownian motion and $t_1, \dots, t_n$ and note the decomposition $X(t_j) = \sum_{k=1}^j X(t_k) - X(t_{k-1})$ and let the sequence of increments $d_k = X(t_k) - X(t_{k-1}), k = 1, \dots, n$. Then, note that $d_k \sim \mathcal{N}(0, t_k - t_{k-1}) \sim \sqrt{t_k - t_{k-1}} \mathcal{N}(0, 1)$. Since increments are independent $d_k \sim \sigma_i Z_i$ where Z_i is a sequence of i.i.d. $\mathcal{N}(0, 1)$ RVs if define $\sigma_i = \sqrt{t_i - t_{i-1}}$. Thus,

$$\vec{X} = \begin{pmatrix} X(t_1) \\ \vdots \\ X(t_n) \end{pmatrix} \sim \begin{pmatrix} d_1 \\ d_1 + d_2 \\ \vdots \\ d_1 + \dots + d_n \end{pmatrix} \sim \begin{pmatrix} \sigma_1 Z_1 \\ \sigma_1 Z_1 + \sigma_2 Z_2 \\ \vdots \\ \sigma_1 Z_1 + \dots + \sigma_n Z_n \end{pmatrix} = \begin{pmatrix} \sigma_1 & 0 & 0 & \dots & 0 \\ \sigma_1 & \sigma_2 & 0 & \dots & 0 \\ \sigma_1 & \sigma_2 & \sigma_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_1 & \sigma_2 & \sigma_3 & \dots & \sigma_n \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_n \end{pmatrix}$$

□

Note. Let's directly compute for $s \leq t$, $\text{Cov}(X(s), X(t))$. We have:

$$\begin{aligned} \text{Cov}(X(s), X(t)) &= \text{Cov}(X(s), X(t) - X(s) + X(s)) \\ &= \text{Cov}(X(s), X(s)) + 0 \\ &= \text{Var}(X(s)) = s \end{aligned}$$

Theorem (Conditioning on a Future Value)

Let $s < t$, then for a Brownian Motion $X(t)$:

$$X(s) \mid X(t) = B \sim \mathcal{N}\left(\frac{Bs}{t}, \frac{s(t-s)}{t}\right)$$

Proof. Directly compute the conditional density of $X(s) \mid X(t) = B$:

$$\begin{aligned} f_{X(s), X(t)=B}(x \mid B) &= \frac{f_{X(s), X(t)}(x, B)}{f_{X(t)}(B)} \\ &= \frac{f_s(x) f_{t-s}(B-x)}{f_t(B)} \\ &= \frac{1}{\sqrt{2\pi \frac{(t-s)}{t}}} e^{-\frac{(x - \frac{Bs}{t})^2}{2 \frac{(t-s)}{t}}} \end{aligned}$$

□

Remark

If (Y, Z) has a bivariate normal distribution then $Y \mid Z = B$ has the normal distribution.

Proof. Best shown directly. Let $\mu_Y = \mathbb{E}(Y)$, $\mu_Z = \mathbb{E}(Z)$, $\sigma_Y^2 = \text{Var}(Y)$, $\sigma_Z^2 = \text{Var}(Z)$, $\rho = \text{Cov}(Y, Z)$, then the explicit form of the bivariate normal joint density is:

$$f_{Y,Z}(y, z) = \frac{1}{2\pi \sigma_Y \sigma_Z \sqrt{1 - \rho^2}} \exp\left(-\frac{1}{2(1 - \rho^2)} \left[\frac{(y - \mu_Y)^2}{\sigma_Y^2} - 2\rho \frac{(y - \mu_Y)(z - \mu_Z)}{\sigma_Y \sigma_Z} + \frac{(z - \mu_Z)^2}{\sigma_Z^2} \right]\right)$$

To compute the conditional density, we must integrate out y to get $f_Z(z)$, thus let's complete the square in y , so that we may easily compute the Gaussian integral. First, for the sake of my sanity I will define $d_Y = y - \mu_Y$, $d_Z = z - \mu_Z$. Then the exponential argument of the joint bivariate normal density is:

$$\begin{aligned} \frac{d_Y^2}{\sigma_Y^2} - \frac{2\rho d_Y d_Z}{\sigma_Y \sigma_Z} + \frac{d_Z^2}{\sigma_Z^2} &= \frac{1}{\sigma_Y^2} \left(d_Y^2 - \frac{2d_Y \rho d_Z \sigma_Y}{\sigma_Z} + \frac{\rho^2 d_Z^2 \sigma_Y^2}{\sigma_Z^2} \right) - \frac{\rho^2 d_Z^2}{\sigma_Z^2} + \frac{d_Z^2}{\sigma_Z^2} \\ &= \frac{1}{\sigma_Y^2} \left(d_Y - \frac{\rho d_Z \sigma_Y}{\sigma_Z} \right)^2 + \frac{d_Z^2}{\sigma_Z^2} (1 - \rho^2) \end{aligned}$$

This allows us to separate out the z -dependence:

$$\begin{aligned} f_{Y,Z}(y, z) &= \frac{1}{2\pi \sigma_Y \sigma_Z \sqrt{1 - \rho^2}} e^{-\frac{1}{2(1 - \rho^2)} \frac{d_Z^2}{\sigma_Z^2} (1 - \rho^2)} e^{-\frac{1}{2(1 - \rho^2)} \frac{1}{\sigma_Y^2} (y - \mu_Y - \frac{\rho d_Z \sigma_Y}{\sigma_Z})^2} \\ f_Z(z) &= \int_{-\infty}^{\infty} f_{Y,Z}(y, z) dy \\ &= \frac{1}{\sigma_Z \sqrt{2\pi}} e^{-\frac{(z - \mu_Z)^2}{2\sigma_Z^2}} \int_{-\infty}^{\infty} \frac{1}{\sigma_Y \sqrt{1 - \rho^2} \sqrt{2\pi}} e^{-\frac{1}{2\sigma_Y^2(1 - \rho^2)} (y - \mu_Y - \frac{\rho d_Z \sigma_Y}{\sigma_Z})^2} dy \\ &= \frac{1}{\sigma_Z \sqrt{2\pi}} e^{-\frac{(z - \mu_Z)^2}{2\sigma_Z^2}} \\ \implies f_{Y|Z}(y \mid z) &= \frac{1}{\sigma_Y \sqrt{1 - \rho^2} \sqrt{2\pi}} e^{-\frac{1}{2\sigma_Y^2(1 - \rho^2)} (y - \mu_Y - \frac{\rho d_Z \sigma_Y}{\sigma_Z})^2} \\ \implies Y \mid Z = B &\sim \mathcal{N}\left(\mu_Y + \frac{\rho \sigma_Y (B - \mu_Z)}{\sigma_Z}, \sigma_Y^2 (1 - \rho^2)\right) \end{aligned}$$

□

Remark (Conditioning on $X(t)$ Yields Gaussian)

The process $X(s) \mid X(t), s \leq t$ is Gaussian

Proof. Note that conditional on $X(t) = B, \{X(t), 0 \leq t \leq 1\}$ is a Gaussian process $\iff (X(t_1), \dots, X(t_n)) \mid X(t) = B \sim MVN$ for all $t_1, \dots, t_n \in [0, 1]$. Let $t_1, \dots, t_n$ be arbitrary. We show that for any $a_1, \dots, a_n \in \mathbb{R}$, $a_1 X(t_1) + \dots + a_n X(t_n) \mid X(t) = B$ has a multivariate normal distribution. Note that because $X(t)$ Brownian motion, it is a Gaussian process and thus $(X(t_1), \dots, X(t_n), X(1))$ has a multivariate normal distribution. Thus, $(a_1 X(t_1) + \dots + a_n X(t_n), X(t))$ is bivariate normal because any linear combination of those two is a linear combination of $X(t_1), \dots, X(t_n), X(t)$ and thus normal, which is exactly the definition of BVN. Thus $a_1 X(t_1) + \dots + a_n X(t_n) \mid X(t) = B$ has a (univariate) normal distribution by previous Lemma (a straightforward computation). This completes the proof, by definition of the MVN distribution. $\square$

Note. Since conditional on a future value, the Brownian motion is still a Gaussian process, and those are characterized by the mean vector and covariance matrix let's find:

$$\mathbb{E}(X_s \mid X_t = B) = \frac{Bs}{t}$$

Now, a very straightforward computation yields the covariance for $s < u \leq t$:

$$\begin{aligned} \text{Cov}(X(s), X(u) \mid X(t) = B) &= \mathbb{E}(X(s)X(u) \mid X(t) = B) \\ &\quad - \mathbb{E}(X(s) \mid X(t) = B) \mathbb{E}(X(u) \mid X(t) = B) \\ &= \mathbb{E}(\mathbb{E}(X(s)X(u) \mid X(u), X(t) = B) \mid X(t) = B) - B^2 su \\ &= \mathbb{E}(X(u) \mathbb{E}(X(s) \mid X(u)) \mid X(t) = B) - B^2 su \\ &= \mathbb{E}\left(X(u)X(u) \frac{s}{u} \mid X(t) = B\right) - B^2 su \\ &= \frac{s}{u} \mathbb{E}(X^2(u) \mid X(t) = B) - B^2 su \\ &= \frac{s}{u} [\text{Var}(X(u) \mid X(t) = B) + B^2 u^2] - B^2 su \\ &= \frac{s}{u} \text{Var}(X(u) \mid X(t) = B) + B^2 su - B^2 su \\ &= \frac{s}{u} \frac{u(t-u)}{t} \end{aligned}$$

Brownian Bridge

This is the special case.

Definition (Brownian Bridge Process)

Let $X(t)$ be a Brownian motion. Then $X(t) \mid X(1) = 0, 0 \leq t \leq 1$ is a Brownian Bridge process.

Proposition (Brownian Bridge Characterization)

By the previous work, we have that Brownian Bridge is a Gaussian with $\mathbb{E}(X_t) = 0, t \leq 1$ and $\text{Cov}(X_s, X_t) = s(1-t), s < t \leq 1$

8.2 Hitting Times, Maxima, Arcsine Law

We include a picture for each major concept, since these are mostly picture proofs (real proofs would require graduate measure theory).

0.0.1 8.2.1) Hitting Time

For now, take $a > 0$.

Definition (Brownian Motion Hitting Time)

$$T^x := \inf\{t > 0 : X(t) = x\}$$

Theorem (Reflection Principle I)

$$\mathbb{P}(T^a \leq t) = 2\mathbb{P}(X_t \geq a)$$

Explanation. We begin backwards with $\mathbb{P}(X_t \geq a)$ and conditioning on $\{T^a \leq t\}$:

$$\mathbb{P}(X_t \geq a) = \mathbb{P}(X_t \geq a \mid T^a \leq t) \mathbb{P}(T^a \leq t) + \mathbb{P}(X_t \geq a \mid T^a > t) \mathbb{P}(T^a > t)$$

In words: what happens to the likelihood the Brownian motion's position at t is above a depending on whether or the process has yet hit t .

Consider $T^a > t$. At time t , we have not yet hit a . By continuity, since Brownian motion starts from $X(0) = 0$, it must go up in net to reach a and jumps are not allowed. Thus, if

we have not yet hit a , we must be below a . If we were above a , Brownian motion must have gone through a some time before t . The only way this could be false (we are above a but never hit $X(t) = a$), is if the function had a jump discontinuity.

Bottom Line: $\mathbb{P}(X_t \geq a \mid T^a > t) = 0$

Now, we will use a symmetry argument. For Brownian motion, at any hitting or stopping time, the process "resets" (a form of Strong Markov). We can consider $X(T)$ to be the new horizontal axis and after the hit it is equally likely to be above or below $X(T)$ (by symmetry since BM is a sum of up/down steps). By symmetry, after the hitting time T^a , it is equally likely that $X_t \geq a$ as $X_t < a$.

$$\implies \mathbb{P}(X_t \geq a \mid T^a \leq t) = \frac{1}{2}$$

□

For this corollary and in general it is helpful to draw a bell curve and use the symmetry of the normal distribution aggressively.

Corollary (Distribution of First Hitting Time)

Since $X_t \sim \mathcal{N}(0, t)$, we have:

$$\mathbb{P}(T^a \leq t) = 2\mathbb{P}(X_t \geq a) = 2 \left(1 - \Phi \left(\frac{a}{\sqrt{t}} \right) \right) = 2\Phi \left(-\frac{a}{\sqrt{t}} \right)$$

We can express this using the integral formula:

$$\mathbb{P}(T^a \leq t) = 2 \int_a^\infty \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} dx$$

Making the substitution $u = \frac{x}{\sqrt{t}}$, so $x = u\sqrt{t}$ and $dx = \sqrt{t} du$:

$$\mathbb{P}(T^a \leq t) = 2 \int_{a/\sqrt{t}}^\infty \frac{1}{\sqrt{2\pi t}} e^{-\frac{u^2}{2}} \sqrt{t} du = 2 \int_{a/\sqrt{t}}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$$

which confirms:

$$\mathbb{P}(T^a \leq t) = 2 \left(1 - \Phi \left(\frac{a}{\sqrt{t}} \right) \right) = 2\Phi \left(-\frac{a}{\sqrt{t}} \right)$$

where Φ is the standard normal CDF.

Note (Probability of Finite Hitting Time).

$$\mathbb{P}(T^a < \infty) = \lim_{t \rightarrow \infty} \mathbb{P}(T^a \leq t) = 2 \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du = 1 < \infty$$

by standard normal density and integral tricks

Note (Expected Hitting Time Infinite). We will show that the expected hitting time of $a > 0$ for Brownian is infinite:

$$\begin{aligned}
\mathbb{E}(T^a) &= \int_0^\infty \mathbb{P}(T^a > t) dt \\
&= \int_0^\infty \left(1 - 2 \int_{a/\sqrt{t}}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du\right) dt \\
&= \int_0^\infty \left(\int_{-a/\sqrt{t}}^{a/\sqrt{t}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du\right) dt \\
&= 2 \int_0^\infty \int_0^{a/\sqrt{t}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du dt \\
&= \frac{2}{\sqrt{2\pi}} \int_0^\infty e^{-\frac{u^2}{2}} \int_0^{a^2/u^2} dt du \quad \text{Switch order of integration, draw region} \\
&= \frac{2}{\sqrt{2\pi}} \int_0^\infty e^{-\frac{u^2}{2}} \frac{a^2}{u^2} du
\end{aligned}$$

As a final step we show that $\int_0^1 e^{-\frac{u^2}{2}} \frac{a^2}{u^2} du$ diverges. Since the function $e^{-\frac{u^2}{2}}$ is a Gaussian bell curve on $[0,1]$ it is decreasing thus $e^{-\frac{u^2}{2}} \geq \sqrt{e} \implies \int_0^1 e^{-\frac{u^2}{2}} \frac{a^2}{u^2} du \geq a^2 \sqrt{e} \int_0^1 \frac{1}{u^2} du = \infty$

It is worth stating a more formal and stronger version of the Reflection Principle:

Theorem (Reflection Principle II)

Let T be a stopping time. Then the process

$$B(t) = \begin{cases} X(t), t \leq T \\ 2X(T) - X(t), t > T \end{cases}$$

is a standard Brownian motion (i.e. it has the same distribution as $X(t)$)

Proof. The type of thing that would be proved in a graduate course, but we can explain it similar to the last version of the reflection principle: after the hitting time T , the distribution of the path of Brownian motion is the same as that of the path of Brownian motion reflected over the horizontal line $X(T) = a$. This is again, the "reset to 0 at stopping time and then it is equally likely to go up or down argument." Again, a picture helps a lot.

So just remember $B_t \sim X_t$

INSERT PICTURE

□

Remark (Symmetry of Hitting Time Distributions)

By symmetry

$$\boxed{T^a \stackrel{d}{=} T^{-a}}$$
$$\implies \boxed{\mathbb{P}(T^a \leq t) = 2 \int_{|a|/\sqrt{t}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du}$$

8.2.2) Maximum of Brownian Motion

Definition (Brownian Motion Maximum)

If $\{X(t), t \geq 0\}$ is a Brownian Motion, then we define the *maximum random variable*

$$M(t) := \max_{s \in [0, t]} X(s)$$

representing the maximum value of Brownian motion over the interval $[0, t]$

Proposition

The following powerful event equality holds:

$$\{M_t \geq a\} = \{T^a \leq t\}$$

by continuity and common sense

One direction is so simple that even an idiot or moron could see it. But if $M_t \geq a$, the process maximum value in the interval is above or equal to a , so since jumps are not allowed we have $T^a \leq t$. Once this easy fact is established the following thm is trivially easy.

Theorem (Maximum of BM Distribution)

$$\boxed{\mathbb{P}(M_t \geq a) = \mathbb{P}(T^a \leq t) = 2\mathbb{P}(X_t \geq a) = 2 \int_{a/\sqrt{t}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du}$$

Theorem (Reflection Principle Consequence)

We can derive the joint distribution of M_t, X_t using reflection principle. For $y < x$:

$$\mathbb{P}(X_t \leq y, M_t > x) = \mathbb{P}(X_t \geq 2x - y)$$

Proof. Since these RV all continuous, under $\mathbb{P}()$ we can be super sloppy with inequality. Also recall $M_t \geq x \iff T^x \leq t$

$$\mathbb{P}(X_t \leq y, M_t > x) = \mathbb{P}(X_t \leq y, T^x < t) = \mathbb{P}(2X(T^x) - X_t \leq y) = \mathbb{P}(2x - y \leq X_t)$$

So this is easy using the 2nd Reflection Principle form, just reflect path after process hits x . $\square$

8.2.3) Zeros

To close off this easy section we discuss the event that Brownian motion takes the value 0.

Definition

We define the event that Brownian motion $\{X(t), t \geq 0\}$ takes on the value at least once in the interval (t_1, t_2) , denoted $\mathbf{0}(t_1, t_2)$:

$$\mathbf{0}(t_1, t_2) = \{X(t) = 0, t \in (t_1, t_2)\}$$

Proposition

$$\mathbb{P}(\mathbf{0}(t_1, t_2) \mid X(t_1) = x) = \mathbb{P}(T^{|x|} \leq t_2 - t_1)$$

Diagram: BM goes from 0 up to $x > 0$ WLOG at t_1 , then must go ack down to 0. Shift vertically to 0, then it must hit $-x$ by $t_2 - t_1$

Theorem (Arcsine Law for Brownian Motion)

Let $\mathbf{0}(t_1, t_2) = \{X(t) = 0, t \in (t_1, t_2)\}$, then $\mathbb{P}(\mathbf{0}(t_1, t_2)) = 1 - \frac{2}{\pi} \arcsin \sqrt{\frac{t_1}{t_2}}$

Proof.

$$\mathbb{P}(T^{|x|} \leq t_2) = 2 \int_x^\infty \frac{1}{\sqrt{2\pi(t_2 - t_1)}} e^{-\frac{u^2}{2(t_2 - t_1)}} du$$

Note in the above we use fact that $T^{-x} \sim T^x$ to take WLOG $x > 0$. By conditioning on $X(t_1) = x$ we obtain:

$$\begin{aligned}\mathbb{P}(\mathbf{0}(t_1, t_2)) &= \frac{1}{\sqrt{2\pi t_1}} \int_{-\infty}^{\infty} \mathbb{P}(\mathbf{0}(t_1, t_2) \mid X(t_1) = x) e^{-\frac{x^2}{2t_1}} dx \\ &= \frac{1}{\sqrt{2\pi t_1}} \int_0^{\infty} 2 \int_x^{\infty} \frac{1}{\sqrt{2\pi(t_2 - t_1)}} e^{-\frac{u^2}{2(t_2 - t_1)}} du e^{-\frac{x^2}{2t_1}} dx\end{aligned}$$

Evaluate integral for result. □

Proposition

For Brownian motion, let $A(t)$ denote the amount of time in $[0, t]$ the process is positive. Then, for $0 < x < 1$,

$$\mathbb{P}(A(t)/t \leq x) = \frac{2}{\pi} \arcsin \sqrt{x}$$

8.3 Variants of Brownian Motion

Throughout X_t or $X(t)$ is a standard Brownian motion, also T^x hitting time as seen above.

8.3.1) Brownian Motion Absorbed at a Value

Definition (Brownian Motion Absorbed at a Value)

$$Z(t) = \begin{cases} X(t) & \text{if } t < T^x \\ x & \text{if } t \geq T^x \end{cases}$$

The delicious red meat of this section is scrumptiously deriving the distribution of this fascinating object!

Theorem

The distribution (CDF) of BM absorbed at a value is for $y < x$:

$$\mathbb{P}(Z_t \leq y) = \frac{1}{\sqrt{2\pi t}} \int_{y-2x}^y e^{-\frac{u^2}{2t}} du$$

and

$$P(Z_t = x) = \frac{2}{\sqrt{2\pi}} \int_{\frac{x}{\sqrt{t}}}^{\infty} e^{-\frac{u^2}{2}} du$$

Proof. Z_t has a discrete part it is $Z_t = x \iff t \geq T^x$ so we have:

$$\mathbb{P}(Z_t = x) = \mathbb{P}(T^x \leq t) = 2\mathbb{P}(X_t \geq x) = \frac{2}{\sqrt{2\pi}} \int_{\frac{x}{\sqrt{t}}}^{\infty} e^{-\frac{u^2}{2}} du$$

Z_t has a continuous part too! This is when we have not yet hit x . For this we must have $T^x > t \iff M_t < x$. Since $y < x$:

$$\begin{aligned} \mathbb{P}(Z_t \leq y) &= \mathbb{P}(X_t \leq y, M_t < x) \\ &= \mathbb{P}(X_t \leq y) - \mathbb{P}(X_t \leq y, M_t \geq x) \\ &= \mathbb{P}(X_t \leq y) - \mathbb{P}(X_t \leq y, M_t > x) \\ &= \mathbb{P}(X_t \leq y) - \mathbb{P}(X_t \geq 2x - y) \quad \text{By consequence of Reflection Principle II} \\ &= \mathbb{P}(X_t \leq y) - \mathbb{P}(X_t \leq y - 2x) \quad \text{right tail = left tail in Gaussian} \end{aligned}$$

□

8.3.2) Brownian Motion Reflected at Origin ($|X_t|$)

Definition

$$Z_t = |X_t|$$

Remark

Distribution is easily obtained by using standard properties of normal distribution.

$$\mathbb{P}(Z_t \leq y) = \mathbb{P}(|X_t| \leq y) = 2\mathbb{P}(X_t \leq y) - 1 = 2 \int_0^y \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} dx - 1$$

and additionally the mean and variance are:

$$\mathbb{E}(Z_t) = \sqrt{\frac{2t}{\pi}}, \text{Var}(Z_t) = (1 - \frac{2}{\pi})t$$

Proof. Given: $X_t \sim N(0, t)$ and $Z_t = |X_t|$

Density of Z_t : $f_Z(z) = \frac{2}{\sqrt{2\pi t}} e^{-\frac{z^2}{2t}}$ for $z \geq 0$

Mean: $\mathbb{E}[Z_t] = \sqrt{\frac{2t}{\pi}}$

Using the density function:

$$\mathbb{E}[Z_t] = \int_0^{\infty} z \cdot f_Z(z) dz = \int_0^{\infty} z \cdot \frac{2}{\sqrt{2\pi t}} e^{-\frac{z^2}{2t}} dz \quad (12)$$

$$= \frac{2}{\sqrt{2\pi t}} \int_0^\infty z e^{-\frac{z^2}{2t}} dz \quad (13)$$

Substitution: Let $u = \frac{z^2}{2t}$, so $du = \frac{z}{t} dz$, which gives us $z dz = t du$
 When $z = 0$, $u = 0$; when $z = \infty$, $u = \infty$

$$\mathbb{E}[Z_t] = \frac{2}{\sqrt{2\pi t}} \int_0^\infty t e^{-u} du = \frac{2t}{\sqrt{2\pi t}} \int_0^\infty e^{-u} du \quad (14)$$

$$= \frac{2t}{\sqrt{2\pi t}} \cdot 1 = \frac{2t}{\sqrt{2\pi t}} = \sqrt{\frac{4t^2}{2\pi t}} = \sqrt{\frac{2t}{\pi}} \quad (15)$$

Variance: $\text{Var}(Z_t) = (1 - \frac{2}{\pi}) t$

First, we compute $\mathbb{E}[Z_t^2]$:

$$\mathbb{E}[Z_t^2] = \int_0^\infty z^2 \cdot f_Z(z) dz = \int_0^\infty z^2 \cdot \frac{2}{\sqrt{2\pi t}} e^{-\frac{z^2}{2t}} dz \quad (16)$$

$$= \frac{2}{\sqrt{2\pi t}} \int_0^\infty z^2 e^{-\frac{z^2}{2t}} dz \quad (17)$$

Substitution: Let $u = \frac{z^2}{2t}$, so:

- $z = \sqrt{2tu}$
- $dz = \frac{\sqrt{2t}}{2\sqrt{u}} du = \frac{\sqrt{t}}{\sqrt{2u}} du$
- $z^2 dz = 2tu \cdot \frac{\sqrt{t}}{\sqrt{2u}} du = \sqrt{2t^3} \sqrt{u} du$

$$\mathbb{E}[Z_t^2] = \frac{2}{\sqrt{2\pi t}} \int_0^\infty \sqrt{2t^3} \sqrt{u} e^{-u} du \quad (18)$$

$$= \frac{2\sqrt{2t^3}}{\sqrt{2\pi t}} \int_0^\infty u^{1/2} e^{-u} du = \frac{2\sqrt{2t^3}}{\sqrt{2\pi t}} \cdot \Gamma(3/2) \quad (19)$$

Since $\Gamma(3/2) = \frac{1}{2}\Gamma(1/2) = \frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{2}$:

$$\mathbb{E}[Z_t^2] = \frac{2\sqrt{2t^3}}{\sqrt{2\pi t}} \cdot \frac{\sqrt{\pi}}{2} = \frac{\sqrt{2t^3}}{\sqrt{2\pi t}} \cdot \sqrt{\pi} \quad (20)$$

$$= \sqrt{\frac{2t^3 \cdot \pi}{2\pi t}} = \sqrt{t^2} = t \quad (21)$$

Final calculation of variance:

$$\text{Var}(Z_t) = \mathbb{E}[Z_t^2] - (\mathbb{E}[Z_t])^2 = t - \left(\sqrt{\frac{2t}{\pi}} \right)^2 \quad (22)$$

$$= t - \frac{2t}{\pi} = t \left(1 - \frac{2}{\pi} \right) = \left(1 - \frac{2}{\pi} \right) t \quad (23)$$

□

Geometric BM

Definition

$$Y_t = e^{X_t}$$

Proposition (Geometric BM Mean and Variance)

$$\begin{aligned} \mathbb{E}(Y_t) &= e^{\frac{t}{2}} \\ \text{Var}(Y_t) &= e^{2t} - e^t \end{aligned}$$

Proof. Recall that for $Z \sim \mathcal{N}(\mu, \sigma)$, the mgf is $\phi_Z(s) = e^{\mu s + \frac{\sigma^2 s^2}{2}}$. We have $X_t \sim \mathcal{N}(0, t)$ so $\phi_{X_t}(s) = \mathbb{E}(e^{sX_t}) = e^{ts^2/2}$ so

$$\mathbb{E}(Y_t) = e^{t/2}$$

Also, for the variance we have:

$$\text{Var}(Y_t) = \mathbb{E}(Y_t^2) - \mathbb{E}(Y_t)^2 = \mathbb{E}(e^{2X_t}) = e^t = e^{2t} - e^t$$

□

This variant of BM is useful if the percent changes (rather than absolute changes) are i.i.d. An example is a stock. If $Y(n)$ is the price of the stock at time n , we know that the percent changes as a fraction $X_n = \frac{Y_n}{Y_{n-1}} \implies Y_n = X_1 X_2 \dots X_n \implies \log Y_n = \sum_{i=1}^n \log(X(n))$, so we may the limit of this process as $n \rightarrow \infty$ as a BM which is the sum of i.i.d. $\log(X_n)$. Thus, Y_n is the exp of a BM.

Proposition (Other Variants)

The following variants are all Brownian Motions

1. **Shift:** $B(t) = X(t + c) - X(c), c > 0$
2. **Scale:** $B(t) = \frac{1}{\sqrt{x}}X(ct), c > 0$
3. **Temporal Inversion:**

$$B_t = \begin{cases} tX(\frac{1}{t}) & t > 0 \\ 0 & t = 0 \end{cases}$$

All are trivially shown except continuity at 0 in (3) which follows from the fact that

$$\lim_{t \rightarrow \infty} \frac{X_t}{t} = 0$$

8.3.4) Integrated Brownian Motion

Motivation: Model price of a commodity at time t , Z_t . Assume that the rate of change of the commodity price $Z'(t)$ follows a BM process rather than the price itself. Think $Z'(t)$ as inflation rate which is a BM.

Definition (Integrated Brownian Motion)

Let $\{X(t), t \geq 0\}$ be a Brownian Motion, then the process $\{Z(t), t \geq 0\}$ defined by:

$$Z(t) = \int_0^t X(s)ds$$

is called *integrated Brownian motion*

Proposition

Z_t is a Gaussian process, but not a Markov process

Proof Sketch. Easy. Use the linear combination definition of MVN. $a_1Z(t_1) + \dots + a_nZ(t_n) =$ the limit of a Riemman sum of $X(s)$ which is Gaussian, thus the limit of Riemman sum of a linear combination of an MVN sequence $b_1X(s_1) + \dots + b_MX(s_M) \implies MVN$. It is not a Markov process because the conditional distribution of a future state given the entire past history and the present depends on the past as it is an integral over the past history. $\square$

Note. Joint process (Z_t, X_t) is Markov

We now compute with abandon the mean and covariance

Proposition (Mean and Covariance of IBM)

$$\begin{aligned}\mathbb{E}(Z_t) &= 0 \\ \text{Var}(Z_t) &= s^2\left(\frac{t}{2} - \frac{s}{6}\right) \quad (\text{clearly})\end{aligned}$$

Proof.

$$\mathbb{E}(Z_t) = \mathbb{E}\left(\int_0^t X(s)ds\right) = \int_0^s \mathbb{E}(X_s) ds = 0$$

Take $s \leq t$

$$\begin{aligned}\text{Cov}(Z_s, Z_t) &= \mathbb{E}(Z_s Z_t) - \mathbb{E}(Z_s) \mathbb{E}(Z_t) \\ &= \mathbb{E}\left(\int_0^s X(u)du \int_0^t X(y)dy\right) \\ &= \mathbb{E}\left(\int_0^t \int_0^s X(u)X(y)dudy\right) \\ &= \int_0^t \int_0^s \mathbb{E}(X(u)X(y)) dudy \\ &= \int_0^t \int_0^s \min(u, y) dudy \\ &= \int_0^s \int_0^u ydydu + \int_0^s \int_u^t udydu \\ &= \frac{1}{6}s^3 + \int_0^s (ut - u^2)du \\ &= \frac{1}{6}s^3 + \frac{1}{2}s^2t - \frac{1}{3}s^3\end{aligned}$$

□

Theorem (Joint Process Distribution)

The joint process $(Z_t, X_t)_{t \geq 0}$ has a bivariate normal distribution with all means 0 and $\text{Cov}(X_t, Z_t) = \frac{t^2}{2}$

Proof. Since both are multivariate normal, clearly the process is bivariate normal (i.e $aX_t + bZ_t$ is normal because $X_t, Z_t = \int_0^s X_t$ and so the linear combination is just the limit of a sum of X's which is Gaussian so the sum is normal). We then proceed to find the covariance (mean is clear). First note:

$$\begin{aligned}\text{Cov}(Z(t), Z(t) - Z(t-h)) &= \frac{1}{3}t^3 - (t-h)^2\left(\frac{t}{2} - \frac{t-h}{6}\right) \\ &= \frac{1}{3}t^3 + \frac{1}{2}t^2h - \frac{t^3}{3} + o(h)\end{aligned}$$

next we have $Z'(t) = \lim_{h \rightarrow 0} \frac{Z_t - Z_{t-h}}{h} = X(t)$ thus $Z(t) - Z(t-h) = X(t)h + o(h)$ (standard approximation of integral over small interval, function value * length of interval + error term). Thus, we have

$$\text{Cov}(Z_t, Z_t - Z_{t-h}) = \text{Cov}(Z_t, hX_t + o(h)) = h\text{Cov}(Z_t, X_t) + o(h)$$

letting $h \rightarrow 0$ implies the result □

Remark

Lastly, we can look at another form of IBM where the percent rate of change of price is Brownian motion:

$$W'(t) = X(t)W(t)$$

Take $W(0) = 1$, then we see that $W(t) = e^{Z_t}$ and that $\mathbb{E}(W_t) = e^{t^3/6}$

$$\frac{dW}{W} = X(t)dt \implies \ln W(t) - \ln W(0) = \int_0^t X(s)ds \implies W_t = e^{Z_t}$$

We use the Geometric BM MGF method. First note that Z_t being MVN means that Z_t normal with mean 0 and $\text{Var}(Z_t) = \frac{t^3}{3}$ and thus we can plug in to the MGF

Summary Table